Counterfactual Generation from Audit Trails in Multi-Agent Negotiation Systems

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Abstract

When autonomous agents negotiate on behalf of human principals, adverse outcomes raise the question of which changes would have altered the result. We formulate counterfactual audit as constrained intervention search in a structural causal model of an agent decision pipeline. An incomplete trace leaves uncertainty about the historical latent state, so an auditor must select one intervention that works across that uncertainty. We seek the least costly changes to evidence, policy checks or negotiation rules that would reverse a decision with a specified level of confidence. Our main result is that missing information can make combined changes essential for a cheap, reliable explanation. For a restricted class of decisions, we show how to select changes that reverse an approval with high probability. Finding the cheapest combination is computationally hard, but a simple search method has guarantees on cost and reliability. Exact affine and box-constrained optimisation references connect the audit margin to adversarial robustness while distinguishing boundary distance, feasible interventions and optimality certificates. Experiments in synthetic negotiation models show how explanations can fail when an audit ignores agents' responses or keeps later messages and decisions fixed after an earlier change. The resulting guarantees hold under the declared model of the interaction and give a basis for explanations from incomplete records.

Keywords

Counterfactual audit, multi-agent negotiation, causal models, posterior uncertainty, submodular cover, adversarial robustness

1 Introduction

With the advent of LLM-based agents as autonomous economic actors, negotiating contracts, executing transactions, and managing portfolios on behalf of human principals, the nature of multi-party coordination is changing profoundly. By dramatically reducing search, contracting, and enforcement costs, these systems approach what has been termed a "Coasean singularity" [39]: flexible, context-dependent workflows that span organisational boundaries and enable coordination patterns that were previously infeasible. But this capability creates an accountability gap. When an agent-mediated decision produces an adverse outcome, neither the principal nor the regulator can readily determine which inputs, disclosures, or policy clauses were decision-critical, nor what changes would have altered the result. The accountability infrastructure must scale with the coordination capability it enables.

A negotiation transcript records what the agents said and did. It can still leave the decisive question unanswered: *what would have had to change for the outcome to be different?* Evidence may pass through several agents, policy checks and verification procedures, while private information and internal choices remain unrecorded. An explanation that works for one possible account of the interaction may fail for another. The auditor needs a change that remains effective across the uncertainty left by the record.

Consider an approved loan whose audit trail leaves open which of two checks was decisive. Under one possible history, withdrawing an evidence item would reverse approval. Under another, overriding a clause verdict would do so. If the history were known, one inexpensive change would suffice. With an incomplete record, neither change alone gives a reliable answer. Insisting that an explanation change only evidence, or only clause verdicts, can force the auditor to use a much more expensive alternative. Our central result makes this gap precise:

> *Missing information can make a combination of changes essential for a cheap, reliable counterfactual, even when every fully known case has a cheapest explanation confined to one stage of the process.*

Moreover, the extra cost of insisting on a single stage has no universal bound (Proposition 6.1).

This observation changes how an audit should be designed. We ask for the least-cost feasible change that would alter the recorded decision with a specified level of confidence; the change is chosen using the available trace. Evidence revisions, internal verdict overrides and protocol changes therefore need to be considered together, with their effects propagated through the agents' decision process. The distinction also gives a useful guarantee: under an explicit condition, restricting attention to one stage preserves a calculable fraction of the original confidence (Section 6.1).

One interpretation is as follows. For a class of prerequisite-based decisions, each possible hidden history determines a set of checks whose failure would reverse the recorded approval. A reliable audit must select changes that reach enough of those possible histories. This turns the problem into a form of set coverage: combining changes extends the histories covered, with diminishing returns. Finding the cheapest combination remains hard, but a simple greedy rule has a provable cost–confidence tradeoff.

2 Related Work

Counterfactual audit of multi-agent systems draws on several research traditions, which we discuss in turn.

- Causal inference and counterfactual identification supply the formal language for counterfactual reasoning and the coupling assumptions on which it rests.
- Counterfactual explanation and algorithmic recourse develop methods for generating decision-changing interventions.
- Adversarial robustness studies the dual problem of minimal decision-changing perturbations.
- Combinatorial optimisation provides the covering arguments and hardness reductions behind our search guarantees.
- Information economics and strategic classification model counterparties who adapt their disclosures to the decision rule.
- Structured transparency designs the information flows between mutually distrusting parties that an audit later examines.
- Verifiable causality analysis addresses the trust problem when audit results must be checked by third parties.

Causal inference and counterfactual identification. Structural causal models give a counterfactual a definite meaning through abduction, action and prediction [36], and every guarantee in this paper is stated relative to a declared model of this kind. The value of a counterfactual depends on a coupling assumption, that is, on how the unobserved state of the recorded interaction carries over to the hypothetical one. Models that agree on every observable quantity can still disagree about this coupling: observational and experimental data in general only bound the probabilities of causation [43], and a Gumbel-max parametrisation is one explicit coupling choice for discrete mechanisms [33]. Section 7 records a positive scalar identification result and two limitations for our setting, and Section 5 states confidence relative to the posterior over latent states under the declared model.

Counterfactual explanation and algorithmic recourse. Counterfactual explanations seek changes associated with a different decision [48], and actionable recourse constrains which changes are available [47]. Causal recourse replaces feature perturbations by interventions [23], and probabilistic recourse addresses imperfect causal knowledge [24]. We adopt the causal formulation and examine layer restrictions when one intervention must succeed across unresolved states of a recorded agent interaction. Tian and Srivastava [44] connect graph explanations to population-level intervention coverage, whereas the uncertainty here concerns a single historical interaction and its intervention layers. Counterfactual reasoning in sequential multi-agent systems also supports attribution. Triantafyllou et al. [46] decompose an action's counterfactual effect through subsequent agent behaviour and state transitions. Our optimisation instead selects a feasible, costed intervention at a required posterior success level. Both tasks rest on the coupling assumption described above.

Adversarial robustness. The minimal counterfactual problem is structurally similar to the adversarial perturbation problem: both seek the smallest input change that flips a decision. Szegedy et al. [42] first demonstrated that neural networks are vulnerable to imperceptibly small perturbations, and subsequent work has developed both attacks (projected gradient descent, Carlini–Wagner [8], DeepFool [30]) and fundamental limits on robustness [12]. That the adversarial distance and the counterfactual margin m are instances of the same optimisation with different cost functions and feasible sets has been made precise in the explanation literature [13, 35]. We use this correspondence (Sections 3.1 and 3.4) in a layered, causally structured perturbation space, where it yields per-layer margins and supplies reference geometry and candidate-generation methods under the declared intervention constraints.

Combinatorial optimisation. The active-ancestor representation of Section 6.2 connects to reachability arguments in influence maximisation [26], and the greedy analysis of Section 6.3 specialises submodular covering [50]. The complexity of counterfactual inference [18] is a different question from optimising one common action across states: in our reduction, evaluating an action is elementary and selection is what remains hard.

Information economics and strategic classification. The distinction between factual and disclosure counterfactuals (Section 8) draws on adverse selection and strategic information provision [2, 41]. Classical unravelling results [16, 29] use assumptions about verifiability, information endowments, preferences, inference from silence and disclosure costs, and the revelation principle [31] concerns implementation of equilibrium outcomes under specified incentive conditions. In machine learning, the same concern appears when decision subjects respond to a decision rule, so that the distribution of observed features becomes endogenous to the rule itself. Hardt et al. [19] formalise this as *strategic classification*, in which agents manipulate their features at a cost to obtain favourable decisions, and Perdomo et al. [37] generalise it to *performative prediction*, in which deployed models shift the data distribution they are evaluated on. Our strategic model declares a utility, an admissible report catalogue and a response rule, so that regret can be measured directly in a finite proving ground. The counterparty's evidence is then the output of a disclosure strategy, and the audit must distinguish counterfactuals over the true state from counterfactuals over the disclosure map. When evidence is treated as exogenous this distinction does not arise. Once evidence is modelled as a response, agent behaviour must be regenerated after a policy change, a requirement that our sequential evaluation makes explicit.

Structured transparency. The disclosure protocols whose conformance our framework audits are an instance of *structured transparency* [45]: information-flow designs that let mutually distrusting parties share, verify and act on sensitive information under explicit bounds, rather than forcing a choice between full disclosure and refusal. Counterfactual audit is the retrospective complement of such designs. Structured transparency constrains which information *may* flow, and the audit determines which flows were decision-critical.

Verifiable causality analysis. Song et al. [40] develop authenticated provenance analysis using a graph accumulator and a verifiable provenance graph. Their work motivates separating the integrity of recorded dependencies from the correctness of hypothetical SCM evaluation. Section 3.5 gives a local evaluation check under declared boundary inputs and mechanism versions. Causal

validity, logging completeness and optimality are separate claims that such a check does not cover.

3 Structural Causal Model for Agent Decisions

We model the agent coordination pipeline as a *structural causal model* (SCM) [36] over the variables of the interaction. Informally, a *causal graph* is a directed acyclic graph whose nodes are the variables of the system and whose edges record direct mechanistic dependence: an edge $X \to Y$ asserts that X is an input to the mechanism that generates Y. While the graph encodes the qualitative causal assumptions (which variables respond to which), the quantitative content is supplied by attaching to each node a *structural equation*, a function producing the node's value from the values of its graph parents together with an exogenous noise term that captures everything outside the model. Acyclicity ensures the system can be evaluated by a single forward pass in topological order (parents evaluated before their children). An *intervention* deletes Y's structural equation (equivalently, severs its incoming edges) and substitutes the fixed value, leaving every other mechanism untouched. This differs from conditioning on the observation $Y = y$, which leaves the incoming dependencies in place. We refer the reader to [38] for an accessible introduction to causal inference and structural causal models.

Formally, an SCM is a tuple $\mathcal{M} = (\mathbf{V}, \mathbf{U}, \mathcal{F}, P_{\mathbf{U}})$, where $\mathbf{V}$ is the set of endogenous variables, $\mathbf{U}$ is the set of exogenous (latent) variables, and $\mathcal{F} = \{f_V\}_{V \in \mathbf{V}}$ is the set of structural equations, with each f_V determining V as a function of its parents $\mathrm{pa}(V)$ in the causal graph $\mathcal{G}$ and the relevant exogenous variables, and $P_{\mathbf{U}}$ is their joint probability law. We assume standard Borel variable spaces and measurable structural equations. The central operation on an SCM is the *intervention*, Pearl's do-operator.

Definition 3.1. A *trace* τ is the observed record of a single execution of an SCM. It specifies the realised values of a subset of the endogenous variables $\mathbf{V}$, including the terminal decision $D = d$. A latent state $\mathbf{u}$ is *compatible* with τ if evaluating the SCM at $\mathbf{U} = \mathbf{u}$ reproduces every recorded value. A trace need not record every endogenous variable or determine the latent state uniquely.

Conditioning on the trace gives a regular conditional posterior $P_{\mathbf{U}}(\cdot \mid \tau)$, which need not factor across mechanisms. Historical counterfactuals retain the same latent state in the factual and intervened worlds.

Definition 3.2. Let $\mathcal{M}$ be an SCM with causal graph $\mathcal{G}$, let $D \in \mathbf{V}$ be a designated outcome variable with observed value d, and fix an exogenous state $\mathbf{u}$ compatible with the observed trace. A *counterfactual intervention* is a set $\delta = \{\mathrm{do}(V_i := v_i')\}_{i \in S}$ for some $S \subseteq \mathbf{V} \setminus \{D\}$, which replaces the structural equations of the variables in S with the prescribed values v_i' and recomputes all downstream variables by forward-propagating through $\mathcal{F}$ in topological order, holding the exogenous variables $\mathbf{U}$ fixed at $\mathbf{u}$. The resulting counterfactual outcome is denoted $d_{\mathcal{M}}(\delta; \mathbf{u})$, or $d_{\mathcal{M}}(\delta)$ when the fixed state is understood.

We say δ is *decision-changing* if $d_{\mathcal{M}}(\delta) \neq d$. In Pearl's three-step procedure (see 36, Chapter 7, §§7.1.2–7.1.3), abduction first conditions the exogenous law on the observed trace τ, giving $P_{\mathbf{U}}(\cdot \mid \tau)$. The intervention step replaces the specified structural equations. For each compatible $\mathbf{u}$, prediction evaluates the intervened model using the same $\mathbf{u}$ as in the factual world. Counterfactual probabilities are obtained by averaging over the posterior, either by exact summation or integration or by repeated posterior draws, each of which gives one counterfactual sample.

The optimisation below first fixes a compatible $\mathbf{u}$ and asks how cheaply an intervention can succeed at that state. When abduction leaves uncertainty, the answer may depend on $\mathbf{u}$. In Section 5, a single intervention is chosen from the observed trace, and its success probability is evaluated across the posterior.

Definition 3.3 (Minimal counterfactual identification). Given an SCM $\mathcal{M}$, an outcome variable D with observed value d, a fixed compatible exogenous state $\mathbf{u}$, a feasible set Δ of interventions, and a cost function $\mathrm{cost}\colon \Delta \to \mathbb{R}_+$ measuring the "size" of an intervention, define the *counterfactual margin* by

$$m = \inf\big\{\mathrm{cost}(\delta) : \delta \in \Delta,\ d_{\mathcal{M}}(\delta) \neq d\big\}, \tag{1}$$

with $\inf \varnothing = +\infty$. A *minimal counterfactual* is a feasible decision-changing intervention δ^* with $\mathrm{cost}(\delta^*) = m$, when such an intervention exists.

If $m < +\infty$, then for every $\varepsilon > 0$ there is a feasible decision-changing *ε-optimal witness* δ_ε with $\mathrm{cost}(\delta_\varepsilon) \leq m + \varepsilon$, even when no minimiser exists. We require cost to be *monotone* under deletion of assignments: if $\delta_A \subseteq \delta$ and both interventions are feasible, then $\mathrm{cost}(\delta) \geq \mathrm{cost}(\delta_A)$. Weighted ℓ_1 norms and sums of nonnegative per-variable charges are additive across disjoint variable sets and hence monotone. A weighted ℓ_∞ norm is monotone but not additively separable, since the cost of a disjoint union is the maximum of its component costs. Where a result requires additivity rather than monotonicity, we say so explicitly.

Definition 3.4 (Layered audit problem). A *layered audit problem* consists of an acyclic SCM $\mathcal{M}$ with a terminal decision $D \in \mathbf{V}$, together with

(1) an ordered partition of $\mathbf{V} \setminus \{D\}$ into layers $\mathbf{V}_1, \ldots, \mathbf{V}_L$ such that no edge of the causal graph points from $\mathbf{V}_j$ into $\mathbf{V}_i$ when $j > i$;
(2) a trace τ;
(3) a catalogue Δ of admissible hard interventions on $\mathbf{V} \setminus \{D\}$, fixed given τ;
(4) a cost $\mathrm{cost}\colon \Delta \to \mathbb{R}_+$ that is *monotone* under inclusion.

An intervention is *pure* if its targets lie in a single layer and *mixed* otherwise.

Write δ_i for the restriction to $\mathbf{V}_i$ and Δ_i for the feasible pure interventions in that layer. The partition is a modelling choice. A finer partition, for example separating evidence from policy, restricts pure search further. In a time-unrolled negotiation, layers may instead represent rounds of the interaction.

Remark 3.5. In a finite-dimensional intervention representation, sufficient conditions for attainment are a nonempty closed successful feasible set and a lower semicontinuous cost with compact sublevel sets on that set: a minimising sequence has a convergent subsequence with a successful limit of minimum cost. For $D(x) = \mathbf{1}[x \geq 0]$ at $x = 1$ and cost $|\delta|$, successful changes satisfy $\delta < -1$. The margin is 1, but the boundary change $\delta = -1$ leaves the decision unchanged and no minimiser exists.

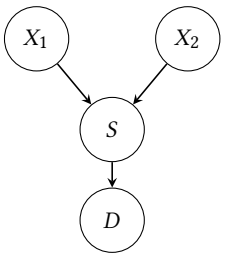
The cost function $\mathrm{cost}(\delta)$ must be chosen to produce meaningful counterfactuals: its design is domain-specific and discussed in detail in Section 3.3. We first illustrate both definitions with a toy example.

Example 3.6 (A simple SCM for a loan decision). Consider a lending scenario with four endogenous variables: *income* $X_1 \in \mathbb{R}_+$, *debt* $X_2 \in \mathbb{R}_+$, *credit score* $S \in (0,1)$, and *decision* $D \in \{\text{approve}, \text{reject}\}$. The structural equations are:

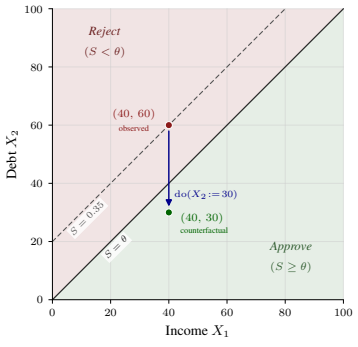
$$X_1 = U_1,\ X_2 = U_2,\ S = \sigma\big((\alpha X_1 - \beta X_2 + U_S)/T\big),$$
$$D = \begin{cases} \text{approve} & \text{if } S \geq \theta, \\ \text{reject} & \text{otherwise,} \end{cases}$$

where $\sigma(x) = (1+\mathrm{e}^{-x})^{-1}$ is the sigmoid function, $\alpha, \beta > 0$ are fixed weights, $T > 0$ is a scale parameter setting the sensitivity of the score to the financial inputs, $\theta \in (0,1)$ is the approval threshold, and U_1, U_2, U_S are independent exogenous variables. Here U_S captures all determinants of the credit score beyond income and debt (payment history, credit utilisation, length of credit history, and measurement noise). It is specific to a given applicant: given the exact observed score and inputs, abduction recovers $U_S = T\log(S/(1-S)) - \alpha X_1 + \beta X_2$, which is then *held fixed* when evaluating the intervention, encoding the assumption that the unmodelled determinants of the score stay the same in the counterfactual world. The causal graph and the resulting decision boundary (for $\alpha = \beta = 1$, $U_S = 0$, $T = 32.3$) are shown in Figure 1.

Suppose we observe a rejected applicant with $X_1 = 40$, $X_2 = 60$, $S = \sigma(-20/32.3) \approx 0.35$, $D =$ reject (with $\theta = 0.5$). The score is rounded for display, and the exact value is consistent with $U_S = 0$. The counterfactual question is: *what is the smallest change to the inputs that would have led to approval?* Since D depends on X_1 and X_2 only through S, we can either intervene on the inputs (an *evidence counterfactual*: e.g., $\mathrm{do}(X_2 := 30)$ raises S above θ) or directly on the score (a *clause counterfactual*: $\mathrm{do}(S := 0.5)$). These are valid candidates. A minimal counterfactual, if attained, minimises $\mathrm{cost}(\cdot)$ over all feasible decision-changing interventions, including mixed interventions when these are permitted.



(a) Causal graph



(b) Decision boundary and counterfactual

Figure 1: Toy SCM of Example 3.6. (a) Causal graph: X_1 (income) and X_2 (debt) determine the credit score S, which determines the decision D. (b) Decision boundary with $\alpha = \beta = 1$, $U_S = 0$, $T = 32.3$.

3.1 The counterfactual margin as optimisation

Example 3.6 admits a closed-form boundary-distance calculation when evidence changes are unrestricted. Since $D = \mathbf{1}[S \geq \theta]$ and $S = \sigma\big((\mathbf{a}^T\mathbf{x} + U_S)/T\big)$ with $\mathbf{a} = (\alpha, -\beta)^T$ and $\mathbf{x} = (X_1, X_2)^T$, the boundary $S = \theta$ corresponds to the hyperplane $H = \{\mathbf{x} : \mathbf{a}^T\mathbf{x} + b = 0\}$ where $b = U_S - T\sigma^{-1}(\theta)$. More generally, let $\mathbf{x}, \mathbf{a} \in \mathbb{R}^p$, $\mathbf{a} \neq \mathbf{0}$, and write $r = \mathbf{a}^T\mathbf{x} + b$ for the factual score residual. First consider the distance to H:

$$m_H = \min_{\delta \in \mathbb{R}^p} \|\delta\| \quad \text{s.t.} \quad \mathbf{a}^T(\mathbf{x}+\delta) + b = 0, \tag{2}$$

where the norm $\|\cdot\|$ encodes the cost, and let δ^*_H denote a minimiser. The affine constraint set is nonempty and closed, and a norm has compact sublevel sets, so this boundary minimum is attained. The calculation temporarily relaxes domain constraints such as $X_1, X_2 \geq 0$ in the toy SCM.

If we use the Euclidean distance ($\|\delta\| = \|\delta\|_2$), the solution can be derived using elementary geometry (see, for example, the lecture notes [28, Chapter 22]):

$$\delta^*_{H,2} = -\frac{r}{\|\mathbf{a}\|_2^2}\,\mathbf{a}, \qquad \|\delta^*_{H,2}\|_2 = \frac{|r|}{\|\mathbf{a}\|_2}. \tag{3}$$

For a cost based on the largest weighted component change, use the *weighted ℓ_∞ norm*

$$\|\delta\|_{w,\infty} = \max_i\ w_i|\delta_i|,$$

where $w_i > 0$ place the coordinates on a declared common cost scale. This norm measures the largest weighted component change rather than the total verification expense or the economic cost of implementing the intervention. We return to the interpretation of cost in Section 3.3.

To solve (2) with this norm, we introduce a Lagrange multiplier $\lambda \in \mathbb{R}$ for the hyperplane constraint and form the Lagrangian

$$\mathcal{L}(\delta, \lambda) = \|\delta\|_{w,\infty} + \lambda\big(r + \mathbf{a}^T\delta\big).$$

To compute the dual, we minimise over δ. The key concept here is the *Fenchel conjugate* of a norm (see, for example, [5]): for any norm $\|\cdot\|$ with dual norm $\|\cdot\|_*$,

$$\inf_\delta \big[\|\delta\| + \mathbf{c}^T\delta\big] = \begin{cases} 0 & \text{if } \|\mathbf{c}\|_* \leq 1, \\ -\infty & \text{otherwise.} \end{cases}$$

The dual of the weighted ℓ_∞ norm is the weighted ℓ_1 norm with reciprocal weights:

$$\|\cdot\|^*_{w,\infty} = \|\cdot\|_{w^{-1},1}, \qquad \|\mathbf{y}\|_{w^{-1},1} = \sum_i \frac{|y_i|}{w_i}. \tag{4}$$

Applying this with $\mathbf{c} = \lambda\mathbf{a}$, the infimum over δ is zero whenever $|\lambda| \cdot \|\mathbf{a}\|_{w^{-1},1} \leq 1$, and $-\infty$ otherwise. The dual problem is therefore

$$\max_\lambda\ \lambda r \quad \text{s.t.} \quad |\lambda| \leq \frac{1}{\|\mathbf{a}\|_{w^{-1},1}}. \tag{5}$$

Strong duality holds because the norm is finite and continuous everywhere and the affine equality is feasible. Put $A = \|\mathbf{a}\|_{w^{-1},1} = \sum_i |a_i|/w_i > 0$. For $r \neq 0$, the dual optimum is $\lambda^* = \mathrm{sign}(r)/A$, and for $r = 0$ we may choose $\lambda^* = 0$. Thus the **unrestricted boundary distance** is

$$m_H = d_{w,\infty}(\mathbf{x}, H) = \frac{|r|}{A} = \frac{|\mathbf{a}^T\mathbf{x} + b|}{\sum_i |a_i|/w_i}. \tag{6}$$

This is the weighted-ℓ_∞ analogue of the Euclidean formula (3): the numerator is the absolute score residual, and the denominator is the corresponding dual norm.

The KKT condition for a boundary optimum is

$$\mathbf{0} \in \partial \|\delta_H^*\|_{w,\infty} + \lambda^* \mathbf{a}.$$

For $\delta \neq \mathbf{0}$, the subdifferential is

$$\partial \|\delta\|_{w,\infty} = \operatorname{conv}\{w_j \operatorname{sign}(\delta_j)\, \mathbf{e}_j : j \in \arg\max_i \, w_i |\delta_i|\},$$

where $\mathbf{e}_j$ is the j-th standard basis vector and conv denotes the convex hull. At zero it is the full dual unit ball:

$$\partial \|\cdot\|_{w,\infty}(\mathbf{0}) = \{\mathbf{y} : \|\mathbf{y}\|_{w^{-1},1} \leq 1\}.$$

For $r \neq 0$, a canonical primal minimiser is

$$(\delta_H^*)_i = \begin{cases} -\operatorname{sign}(r) \operatorname{sign}(a_i)\, m_H / w_i, & a_i \neq 0, \\ 0, & a_i = 0. \end{cases}$$

Indeed, every boundary-feasible δ satisfies $|r| = |\mathbf{a}^T \delta| \leq A \|\delta\|_{w,\infty}$, and this construction attains equality. For $r \neq 0$, equality forces the displayed values on all nonzero-normal coordinates and coordinates with $a_i = 0$ may vary freely within $|\delta_i| \leq m_H / w_i$. Consequently the boundary minimiser is unique if and only if every $a_i \neq 0$. If $r = 0$, the unique minimum-norm perturbation is instead $\delta_H^* = \mathbf{0}$, regardless of zero coordinates in $\mathbf{a}$.

Boundary distance versus a decision flip. For the threshold convention $D = \mathbf{1}[r \geq 0]$ and unrestricted changes, m_H equals the infimum cost of a decision flip in both directions. If $r < 0$, the boundary itself is approved, so δ_H^* attains that infimum. If $r > 0$, rejection requires $r + \mathbf{a}^T \delta < 0$: the infimum is not attained, but $(1+\eta)\delta_H^*$ flips the decision at cost $(1+\eta) m_H$ for every $\eta > 0$. If $r = 0$, the margin is zero and again unattained.

With a restricted evidence feasible set Δ_E and the same norm cost, the evidence-only margin satisfies $m_E \geq m_H$, possibly with $m_E = +\infty$. Immutable coordinates, one-sided changes, bounds, integrality and document-consistency constraints can prevent the unrestricted crossing. Equality requires a feasible successful boundary minimiser or successful feasible changes approaching its cost. Under linear domain constraints, the reject-to-approve problem can be solved using linear programming: minimise t subject to those constraints, $-t/w_i \leq \delta_i \leq t/w_i$ and $r + \mathbf{a}^T \delta \geq 0$. The reverse direction requires a strict crossing and any prescribed positive score buffer defines a different, stronger target.

Remark 3.7 (Relation to adversarial search). Counterfactual and adversarial perturbation problems share an optimisation structure [13, 35]. For example, DeepFool [30] uses local linearisation and boundary projection to propose perturbations. Such proposals require domain-feasibility checks, an actual threshold crossing, and propagation through the declared causal mechanisms.

3.2 Causal Graph Structure for Agent Coordination

We now apply this framework to the richer structure of agent coordination, using the following endogenous variables. This graph represents a controlled decision-stage audit: the recorded policy and transcript are held fixed unless explicitly intervened on. Earlier disclosure effects require the temporal model of Section 8.

- $\mathbf{E} = (E_1, \ldots, E_n)$ denotes the evidence package submitted by the counterparty (financial metrics, verification documents, bank feed data);
- π denotes the local agent's policy state (hard constraints, soft preferences, escalation rules);
- $\mathbf{N} = (N_1, \ldots, N_r)$ denotes the negotiation transcript (proposals, counter-proposals, clarification requests);
- $\mathbf{C} = (C_1, \ldots, C_k)$ denotes the satisfaction verdicts for the active contract and policy clauses (e.g., disclosure requirements and exposure limits);
- $\phi \in \Phi$ denotes the selected protocol. It specifies how agents negotiate, verify evidence and trigger fallbacks;
- D denotes the terminal decision (approve, reject, counter, refer).

Evidence, policy and transcript together form the input layer. Clause verdicts, protocol and decision form the three downstream layers. The exogenous variables $\mathbf{U}$ capture hidden counterparty state, unobserved risk factors, and noise.

The structural equations take the form:

$$C_j = g_j(\mathbf{E}, \pi, \mathbf{N}; U_{C_j}), \quad j = 1, \ldots, k, \tag{7}$$

$$\phi = h(\pi, \mathbf{C}, \mathbf{N}; U_\phi), \tag{8}$$

$$D = q(\mathbf{C}, \phi, \mathbf{E}; U_D), \tag{9}$$

where g_j checks whether clause j is satisfied given the evidence and negotiation context, h selects the protocol given the policy and clause status, and q produces the terminal decision. Formally, a clause π_j specifies inputs x_j from $(\mathbf{E}, \mathbf{N})$, parameters ω_j, and a test $\psi_j(x_j; \omega_j)$ in a finite verdict set such as {pass, fail, insufficient}. Its implemented verdict C_j is the output of g_j and can differ from that test when the mechanism misreads its inputs. Changing ω_j is a policy intervention, whereas overriding C_j replaces only the verdict.

Figure 2 groups the causal graph $\mathcal{G}$ into layers. Each band shows the variables produced at that layer and the inputs read by its mechanism. The input badges retain dependencies that skip layers: evidence enters the decision directly, and clause verdicts enter both protocol selection and the decision. Policy and the shared negotiation state enter clause checks and protocol selection. The vertical rail denotes evaluation order.

In a sequential interaction, one possible representation of the trace in Definition 3.1 is $\tau = (s_0, a_0, s_1, a_1, \ldots, s_T, d)$, where s_t contains the recorded shared-state fields at step t, a_t is the recorded action, and $d \in \mathcal{D}$ is the terminal decision. Other state components and intermediate computations may be unrecorded. The trace supplies observed variable values, while the structural equations (7)–(9) specify their generating mechanisms. Together they constrain the factual execution but need not determine it uniquely. With a declared exogenous law they support abduction, and fixing a compatible exogenous state then permits the counterfactual evaluations below.

Example 3.8 (A concrete lending audit). Consider a borrower agent requesting a loan from a lender agent. Table 1 gives one possible representation of every layer of Figure 2. The records could be stored as JSON objects or database rows. The SCM concerns their named fields and dependencies rather than any particular storage format. We audit a checkpoint after the stated messages

Table 1: Concrete records for the lending audit in Example 3.8. Amounts are illustrative. Monthly quantities use GBP/month and the requested principal uses GBP. Protocol identifiers refer to the procedures described in the example.

Component	Example representation and factual contents
Evidence **E**	A record with named fields: `income` = 3000, `committed` = 2600, `principal` = 12000, `instalment` = 500, `account_blocked` = `false`. Numeric evidence items carry a value, unit, observation period and source reference. The income and committed-outgoings items cite a bank record `B17`, and an attestation accompanies that record.
Policy π	A versioned configuration `loan-policy/v3`: maximum principal 15000; a valid bank attestation is required; a new instalment must fit within income less committed outgoings; review is permitted with consent.
Transcript **N**	An ordered list of message records, each with sequence number, sender, recipient, message type and payload. Here the lender asks whether review is acceptable, and the borrower replies: "I prefer an automatic decision, but consent to referral for review."
Clauses **C**	Named Boolean verdicts $(C_{\text{doc}}, C_{\text{limit}}, C_{\text{pay}}) = (1, 1, 0)$. A log entry for each verdict can additionally retain the checker version, evidence references and a reason, such as "available 400; required 500."
Protocol ϕ	A catalogue identifier and version, here `automatic/v1`. The alternative is `review/v1`. Each names a fixed procedure with specified checks, messages and stopping conditions.
Decision D	The categorical outcome `reject`. A separate explanatory payload records the failed affordability clause. In this example D denotes the category, so changing only a reason string does not count as a decision flip.

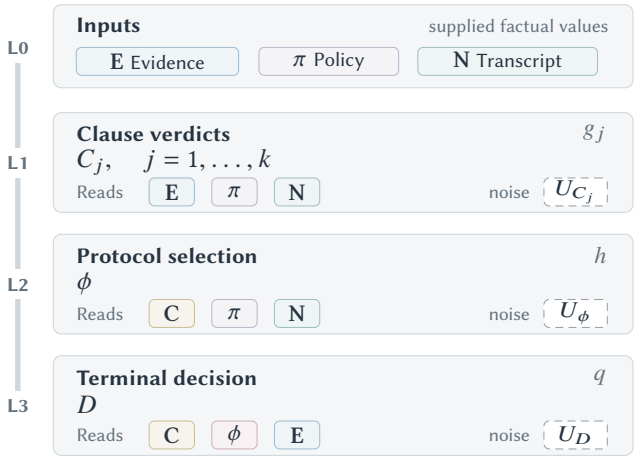


Figure 2: Layered view of the decision-stage SCM. The rail indicates evaluation order, and "Reads" badges name each mechanism's endogenous inputs, including dependencies that skip layers. Dashed badges denote exogenous inputs, whose joint law need not factor.

and evidence have been received. All mechanisms in this example are deterministic, so the exogenous variables can be taken to be constant.

Clause checks. The mechanism g_{doc} checks the bank attestation under the required verification rule and returns one for `B17`. The other two checks are

$$C_{\text{limit}} = \mathbf{1}\{E_{\text{principal}} \le \pi_{\max}\},$$
$$C_{\text{pay}} = \mathbf{1}\{E_{\text{income}} - E_{\text{committed}} \ge E_{\text{instalment}}\}.$$

Thus the principal satisfies the 15000 limit, but the available monthly amount is $3000 - 2600 = 400$, below the proposed instalment of 500. The evidence record, the rule being applied and the resulting verdict are distinct objects. A valid attestation shows that the stated verification condition was met, not that the figures are economically true.

Protocol selection. The selector h chooses `review/v1` if policy permits review, the transcript records consent, and either the document check failed or the borrower requested review. Otherwise it chooses `automatic/v1`. The factual document check passes and the borrower prefers an automatic decision, so h selects `automatic/v1`. A protocol is more than an outcome label: the two catalogue entries specify the following procedures.

- `automatic/v1`: check the account-block flag and, if blocked, emit a rejection and stop. Otherwise read the clause verdicts, emit approval if all pass and rejection otherwise, then stop.
- `review/v1`: check the same block flag and, if blocked, emit a rejection and stop. Otherwise assemble a packet containing the evidence references and clause verdicts, hand it to the reviewing agent, emit a referral and stop this automated stage.

The selector chooses the procedure and q evaluates it. Abbreviating the two protocol identifiers as auto and review, its decision rule is

$$D = \begin{cases} \text{reject}, & E_{\text{blocked}} = 1, \\ \text{refer}, & E_{\text{blocked}} = 0,\ \phi = \text{review}, \\ \text{approve}, & E_{\text{blocked}} = 0,\ \phi = \text{auto},\ \bigwedge_j C_j = 1, \\ \text{reject}, & \text{otherwise.} \end{cases}$$

Here E_{blocked} abbreviates the `account_blocked` field. The factual outcome is rejection. The block check also illustrates why Figure 2 retains a direct evidence-to-decision dependency: q reads that field even though it is not one of these three clause verdicts.

Three different audit interventions. Replacing the income item by a consistent, attested record with value 3100 makes $C_{\text{pay}} = 1$, and recomputing h and q gives approval. Instead overriding only $C_{\text{pay}} := 1$ also gives approval, but leaves the recorded income at 3000: this tests the verdict's causal role without changing the underlying evidence. Finally, overriding $\phi :=$ `review/v1` leaves $\mathbf{C} = (1, 1, 0)$ and produces a referral. Review is permitted and consented to in this case, so that route can be included in the intervention catalogue

even though h did not select it. A referral is a change of decision category, not an approval. Which of the three interventions is cheapest depends on the feasibility and cost specification introduced below.

The evidence replacement includes consistent supporting references and attestations, rather than a numerical edit that invalidates its source record. The transcript is held fixed at the checkpoint. Subsequent review messages and their eventual lending outcome belong to the temporal extension in Section 8, not to this static graph.

3.3 Minimal Counterfactual Identification in the Agent Setting

We now instantiate the general minimal counterfactual problem (Definition 3.3) on the agent coordination graph. The SCM determines the *propagation rule*, while a separately declared intervention catalogue and domain constraints determine the *feasible set*. A counterfactual intervention δ fixes the values of the intervened variables and recomputes all downstream variables by forward-propagating through the structural equations in topological order. Variables that are not descendants of the intervention set in $\mathcal{G}$ remain unchanged. The counterfactual decision is

$$d_{\mathcal{M}}(\delta) = q\big(\mathbf{C}^{(\delta)},\ \phi^{(\delta)},\ \mathbf{E}^{(\delta)};\ U_D\big), \tag{10}$$

where $\mathbf{E}^{(\delta)}$, $\mathbf{C}^{(\delta)}$, and $\phi^{(\delta)}$ are obtained by applying δ and propagating. The minimal counterfactual problem (1) becomes a constrained optimisation over the graph:

$$\begin{aligned} m = \inf_{\delta \in \Delta}\ & \mathrm{cost}(\delta) \\ \text{s.t. } & q\big(\mathbf{C}^{(\delta)}, \phi^{(\delta)}, \mathbf{E}^{(\delta)}; U_D\big) \neq d, \end{aligned} \tag{11}$$

where the feasible set Δ encodes actionability constraints (e.g., the policy π may be fixed, or certain evidence items may be unmodifiable). The convention $\inf \varnothing = +\infty$ and the distinction between a minimiser and an ε-optimal witness remain as in Definition 3.3. The graph structure enters in two ways: through the propagation rule (10), which determines how each candidate intervention affects the decision, and through the layer structure, which enables the decomposition we establish in the next subsection.

The design of cost determines what the reported margin measures, and verification expense, economic action cost and plausibility are distinct criteria. An audit may choose a verification-cost proxy, with variables normalised to declared units and weights calibrated to per-unit checking effort. A weighted ℓ_∞ cost then measures the largest component cost, whereas total effort calls for an additive model when expenses are additive, including any fixed verification charges. Neither choice measures how costly it is for the counterparty to implement a change.

An ℓ_1 penalty can encourage sparse changes, but sparsity does not ensure plausibility, so domain restrictions and any plausibility criterion must be stated separately. Discrete assignment costs and continuous costs can be combined on a declared common scale, provided the resulting scalarisation satisfies the cost assumptions used below.

Remark 3.9 (Mixed modalities and coupled constraints). Evidence can contain continuous quantities, categorical documents and structured text. Clause verdicts and protocols are usually discrete. A continuous proposal method therefore covers only part of the catalogue. The ambient representation may be a product, but actionability, accounting identities and protocol compatibility can couple the coordinates, so the feasible subset need not factor by layer. A pure evidence subproblem can itself mix modalities. The layer bounds below use only feasible-set inclusion, whereas exactness also requires the replacement property stated there.

3.4 Decomposition of Counterfactual Interventions

The layered structure of the causal graph (Figure 2) organises the optimisation problem (11) by layer. Partition the endogenous variables into three intervenable layers, numbered from zero in this pipeline example, and a terminal stage: $L_0 = \{\mathbf{E}, \pi, \mathbf{N}\}$ (roots), $L_1 = \{C_1, \ldots, C_k\}$ (clauses), $L_2 = \{\phi\}$ (protocol), and $L_3 = \{D\}$ (decision). We never intervene on D itself, as D is the quantity we wish to change. Every intervention $\delta \in \Delta$ then decomposes uniquely as a disjoint union $\delta = \delta_0 \cup \delta_1 \cup \delta_2$, where δ_i acts on variables in L_i. We call δ *pure* if only one component is non-empty, and *mixed* otherwise. Pure interventions carry the interpretable audit labels:

(1) **Evidence counterfactuals**: δ_0 acting on the evidence package $\{E_1, \ldots, E_n\}$, that is, changes to the facts that would alter clause satisfaction. Interventions on the policy π, which also lives in L_0, are the policy counterfactuals of Section 8 and are typically excluded from Δ in a factual audit. Interventions on the transcript $\mathbf{N}$ are handled in the same way as evidence.
(2) **Clause counterfactuals**: δ_1 acting on $\{C_1, \ldots, C_k\}$, that is, controlled overrides of clause verdicts.
(3) **Protocol counterfactuals**: δ_2 acting on ϕ, a different protocol selection that would change the decision under the same clauses.

For each layer i, define the *layer-restricted margin*

$$m_i = \inf_{\delta_i \in \Delta_i} \mathrm{cost}(\delta_i) \quad \text{s.t.} \quad d_{\mathcal{M}}(\delta_i) \neq d, \tag{12}$$

where $\Delta_i \subseteq \Delta$ denotes the feasible pure layer-i interventions. We set $m_i = +\infty$ if no pure layer-i intervention flips the decision, and write δ_i^* for a minimiser when one exists.

We say that an instance satisfies *layer-local pivotality* if every decision-changing intervention $\delta = \delta_0 \cup \delta_1 \cup \delta_2 \in \Delta$ has a component $\delta_i \in \Delta_i$ that is feasible and decision-changing on its own, with all non-intervened mechanisms propagated at the same exogenous state.

Proposition 3.10. *Let the SCM be given by* (7)–(9) *and let the cost be monotone in the sense of Definition 3.3.*

(i) $m \leq \min(m_0, m_1, m_2)$.
(ii) *Under layer-local pivotality,* $m = \min(m_0, m_1, m_2)$. *If moreover* $m < +\infty$, *then a minimiser of any layer* i *with* $m_i = m$ *is a global minimiser, and pure* ε*-optimal witnesses exist for every* $\varepsilon > 0$.

Proof. *(i)* Each Δ_i is a subset of Δ, and the constraint in (12) is the constraint of (11) restricted to Δ_i. Hence every feasible point of a layer-restricted problem is feasible for the global problem, and the global infimum is at most $\min(m_0, m_1, m_2)$.

(ii) Let $\delta \in \Delta$ be any decision-changing intervention and let δ_i be a feasible decision-changing component, which exists by layer-local pivotality. Then $\delta_i \in \Delta_i$ is feasible for the layer-i problem (12), so $\mathrm{cost}(\delta_i) \geq m_i \geq \min(m_0, m_1, m_2)$. By monotonicity of the cost, $\mathrm{cost}(\delta) \geq \mathrm{cost}(\delta_i)$. Taking the infimum over decision-changing δ gives $m \geq \min(m_0, m_1, m_2)$, and combined with (i) this gives equality, also when the successful set is empty. If $m < +\infty$, at least one layer has $m_i = m$, since there are finitely many layers. A minimiser for that layer, when it exists, is a pure global minimiser. Without attainment, the definition of the infimum supplies pure ε-optimal witnesses in that layer for every $\varepsilon > 0$.

It remains to record how the constraint $d_{\mathcal{M}}(\delta) \neq d$ is evaluated. Since the structural equations are known (or learned), we apply Pearl's submodel construction [36, Definition 7.1.2]: the intervention replaces the structural equations of the intervened variables with the prescribed values, and all downstream variables are recomputed by forward propagation through the remaining (unmodified) equations, with the exogenous variables held at their abducted values. For evidence counterfactuals ($i = 0$), we set $\mathbf{E}^{(\delta_0)}$, recompute $\mathbf{C}^{(\delta_0)}$ via (7), then $\phi^{(\delta_0)}$ via (8), and finally $D^{(\delta_0)}$ via (9). For clause counterfactuals ($i = 1$), we fix $\mathbf{C}^{(\delta_1)}$ directly and recompute ϕ and D. For protocol counterfactuals ($i = 2$), we fix $\phi^{(\delta_2)}$ and recompute D only. □

When (ii) applies and $m < +\infty$, the audit can report the layer of a pure minimiser or ε-optimal witness as a structured explanation: the decision would have changed if evidence E_i had taken the value e_i', if clause C_j had been satisfied, or if protocol ϕ' had been selected. In general, the audit reports the per-layer components of the returned intervention, its cost and any established optimality gap, together with the pure-layer margins (m_0, m_1, m_2). If $m = +\infty$, no feasible intervention changes the decision.

Proposition 3.11 (Approximate pure replacement). *Suppose $1 \leq \kappa < \infty$ and every successful $\delta \in \Delta$ has a successful feasible pure replacement γ with $\mathrm{cost}(\gamma) \leq \kappa\,\mathrm{cost}(\delta)$. Then*

$$m \leq \min_i m_i \leq \kappa m.$$

In particular, $\kappa = 1$ gives exactness without requiring the replacement to be a component of the original intervention.

The derivation is given in Appendix A.

Remark 3.12 (Exactness regimes and unbounded gaps). Layer-local pivotality holds if only pure interventions are admissible. Another sufficient regime is $D = \bigwedge_j C_j$, with clauses depending only on roots, hard verdict overrides, no other direct inputs to D, and feasibility preserved when taking components. In the approve-to-reject direction, a failed clause is either overridden to zero or made zero by the root component alone. Thus one feasible component flips the decision. This argument does not extend automatically to protocol-dependent rules $q(\mathbf{C}, \phi, \mathbf{E})$.

Without a replacement condition, the gap can be arbitrarily large. Let the factual state be $X = C_1 = C_2 = 0$, with $C_1 = X$, $C_2 = 0$ and $D = C_1 \wedge C_2$. Permit $X := 1$ at cost 1, $C_1 := 1$ at cost $M > 1$, and $C_2 := 1$ at cost 1, with additive costs and all subsets feasible. The mixed intervention $\{X := 1, C_2 := 1\}$ costs 2, while the cheapest pure intervention overrides both clauses at cost $M+1$, and root-only search cannot succeed at all. The pure/global ratio $(M + 1)/2$ is unbounded.

Component feasibility is essential even when isolated components would causally succeed. Let $D = \mathbf{1}[X \geq 1 \text{ or } C = 1]$, factual $X = C = 0$, and permit only the empty intervention, $\{X := 1, C := 1\}$ at cost 2, and $\{X := 2\}$ at cost 3. The mixed components would each flip D, but neither is feasible alone, so the global and pure margins are 2 and 3. The graph alone cannot exclude this case.

Remark 3.13 (Adversarial robustness per layer). For an evidence audit holding policy and transcript fixed, write $m_E = m_0$, $m_C = m_1$, and $m_\phi = m_2$ for the layer-restricted margins (12) of the evidence, clause, and protocol layers. The overall counterfactual margin satisfies $m \leq \min(m_E, m_C, m_\phi)$, with equality under the layer-local pivotality condition of Proposition 3.10(ii), but the triple (m_E, m_C, m_ϕ) is more informative than any single number: a decision that is robust to evidence changes (m_E large) but fragile with respect to protocol selection (m_ϕ small) has a different risk profile from one that is fragile at the evidence layer. These margins are diagnostics for a restricted class of interventions and say nothing about robustness to model changes or to uncertainty about hidden state.

3.5 Verification of Counterfactual Evaluation

An audit certificate must separate record integrity, evaluation correctness, causal-model assumptions, and optimality. Authenticated provenance systems such as vCause [40] motivate commitments to logs and their dependency structure. Authenticating a recorded edge does not establish that it is a correct causal mechanism, or that the logger recorded all relevant events. We assume that the mechanism versions, factual boundary state, intervention and required exogenous state are fixed and available to an evaluator, and treat their authenticity as a separate trust assumption.

Proposition 3.14. *Let $\mathcal{M}$ be a finite acyclic SCM evaluated at a fixed exogenous state $\mathbf{u}$, and let δ be a feasible intervention that overrides the variables in S. Let W consist of D, the variables in S, and every variable on a directed path from S to D, and call a parent of $W \setminus S$ that lies outside W a* boundary input. *Suppose an evaluator holds the mechanism f_v of every $v \in W \setminus S$, authenticated factual values of the boundary inputs, and a proposed value $\hat{x}_v$ for every $v \in W$. If $\hat{x}_v$ equals the value prescribed by δ for every $v \in S$, and $\hat{x}_v = f_v(\hat{x}_{\mathrm{pa}(v)}; \mathbf{u})$ for every $v \in W \setminus S$, with boundary inputs taking their authenticated values, then $\hat{x}_D = d_{\mathcal{M}}(\delta; \mathbf{u})$.*

Proof. A boundary input is not a descendant of S, since otherwise it would lie on a directed path from S to D and hence in W. Its factual value therefore equals its value under δ. Proceed through W in topological order. Overridden variables carry their prescribed values. For $v \in W \setminus S$, if every parent carries its correct value under δ, the local condition forces $\hat{x}_v$ to be correct as well. Induction gives $\hat{x}_D = d_{\mathcal{M}}(\delta; \mathbf{u})$. □

The complete interface of each mechanism matters. Since q reads evidence and clause values directly in addition to the protocol, and g_j and h also read π and $\mathbf{N}$, an evidence intervention can require checking every downstream mechanism, and checking only the evidence and clause records does not establish a terminal flip. The work of ordinary replay is the sum of the evaluated mechanism

costs plus graph traversal, not a function of the number of semantic layers alone. This proposition is independent of the optimisation bound in Proposition 3.10.

If local checks are replaced by cryptographic protocols with soundness errors ϵ_j valid under the chosen composition, an incorrect terminal computation can be accepted only after a local soundness or authentication failure. A union bound, which does not require the failures to be independent, gives error at most $\sum_j \epsilon_j$ plus the authentication failure probability. GKR [15] and SNARKs [17, 34] are possible implementation routes, but their complexity depends on the computation actually encoded, and four semantic layers need not form an arithmetic circuit of depth four. In the experiments we implement local replay checks rather than a cryptographic protocol.

Two limits of such checks should be kept in mind. A proof of surrogate evaluation certifies the surrogate, not the opaque model whose behaviour it approximates, and a commitment to an input/output pair does not prove honest execution. Reproducible stochastic replay also needs immutable code, model and state artifacts, with noise coupled by mechanism and time, because a global seed can assign different draws once the control flow changes. Repeatability of hosted services is a separate question. More generally, neither a valid replay nor a cryptographic evaluation proof establishes the truth of the evidence, causal identification, policy authorisation or minimality of the intervention cost, which needs the separate optimisation bound of the next section.

4 Computational Approach

A search procedure proposes interventions, and a common evaluator checks feasibility and recomputes the terminal decision. For a finite catalogue, exhaustive enumeration gives the global optimum or establishes infeasibility. For the continuous affine case, linear programming and the box support calculation below give reference values. Gradient proposals, finite grids and learned generators generally provide feasible upper bounds only.

Proposition 4.1 (Cost-ordered search certificate). *Let the feasible catalogue Δ be finite, and let its candidates be evaluated exactly in nondecreasing order of cost.*

(i) *The first successful candidate attains m.*
(ii) *If the enumeration stops before a success, the cost of the first unevaluated candidate is a lower bound on m.*
(iii) *If no candidate succeeds, $m = +\infty$.*

The derivation is given in Appendix A.

This is a reference algorithm rather than a scalability result, since constructing, filtering and sorting the catalogue can itself be exponential in the number of intervention targets. We record these costs separately from candidate evaluations. A successful heuristic candidate of cost U and an independently certified bound $L \le m$ together give $L \le m \le U$. The gap $U - L$ is meaningful only for a feasible, successful candidate, and abstention is not proof of infeasibility.

Exact search at bounded structural width. Finite interventions can also be encoded without listing every combined assignment. For each eligible variable, introduce a selector that either retains its structural equation or forces one of its permitted values. Add factors enforcing structural consistency at a fixed compatible $\mathbf{u}$, feasibility, and $D \neq d$, together with factors whose sum is the intervention cost. Infeasible assignments have cost $+\infty$. The resulting primal graph joins any two variables appearing in the same factor and includes selectors and auxiliary variables as well as SCM nodes.

Proposition 4.2 (Exact optimisation at bounded width). *Let the factor encoding above be given explicitly, with maximum domain size q, together with a tree decomposition of its primal graph of width w, and let N be the total input size, including the decomposition and the bit lengths of the rational costs. Then m, together with a minimiser when $m < +\infty$ and a certificate of infeasibility otherwise, can be computed in time $q^{w+1}\,\mathrm{poly}(N)$.*

Proof. Assign each factor to a bag containing its scope. At each bag, a table records the least cost of extending a bag assignment into its descendant subtree. Sum the factors assigned to the bag and the child messages, then minimise over variables absent from the parent separator. The running-intersection property makes the subtrees conditionally separable given the bag assignment. Induction from the leaves establishes the invariant, and minimising the root table gives the global optimum. Backpointers recover a witness when the optimum is finite. Each bag has at most $w + 1$ variables and hence at most q^{w+1} assignments. Processing all factors and messages has the stated complexity. This is standard min-sum elimination [10] applied to the audit encoding. □

The width of the displayed causal DAG alone is not the relevant quantity, since coupled feasibility and high-arity cost factors must be included, and finding the decomposition is a separate task. The result holds at a fixed $\mathbf{u}$ and does not exchange posterior averaging with minimisation over interventions. A finite-scenario posterior encoding would need shared selectors and the probability constraint, whose contribution to the width must also be counted.

Feasibility-aware pruning. A non-ancestor of D does not affect its structural evaluation, but it can be needed to satisfy a joint constraint. For example, changing an accounting field X may require an accompanying certificate field N, even if N has no edge to D. Removing N can destroy feasibility. Restricting the search to ancestors preserves the optimum only if deleting the other assignments preserves feasibility and does not increase cost. Otherwise prune irrelevant computation while retaining the feasibility obligations.

Continuous boxes. For a rejected affine instance ($r < 0$), let $\ell_i \le \delta_i \le u_i$ with $\ell_i \le 0 \le u_i$ and set

$$H(t) = \sum_{a_i > 0} a_i \min(u_i, t/w_i) + \sum_{a_i < 0} a_i \max(\ell_i, -t/w_i).$$

The exact weighted-ℓ_∞ margin is $m_{\mathrm{box}} = \inf\{t \ge 0 : H(t) \ge -r\}$. Indeed, $H(t)$ is the maximum score gain over the box intersected with the cost ball. It is continuous, nondecreasing and piecewise affine, so its breakpoints determine the infimum. For an approved instance, reverse the score direction and require strict crossing: if the maximal gain only reaches the boundary, no flip is feasible. Otherwise report the infimum and a successful ε-optimal witness separately.

Learned proposals. A generator G_θ may be trained with a cost and failure penalty, for example

$$\mathcal{L}(\theta) = \mathbb{E}_{\tau\sim\mathcal{T}}\left[\mathrm{cost}(G_\theta(\tau)) + \mu\mathbf{1}\{d_{\mathcal{M}}(G_\theta(\tau)) = d\}\right]. \tag{13}$$

Here $\mathcal{T}$ is a training distribution separated from evaluation by complete episode. For finite μ, a zero-cost invalid proposal can beat a valid one costing more than μ, so the objective guarantees neither validity nor minimality, and such proposals still need feasibility checks, replay and a separately justified cost bound. Learned search and large-scale LLM evaluation are future extensions of the reference suite below.

5 Counterfactual Confidence and Robustness

A returned audit should state its intervention semantics, validity evidence, cost interval, and epistemic status. At a fixed compatible exogenous state, replay checks validity under the declared model. When the trace leaves latent uncertainty, define the model-conditional posterior flip probability

$$\rho_{\mathcal{M}}(\delta \mid \tau) = \Pr_{\mathbf{U}\sim P_{\mathbf{U}}(\cdot\mid\tau)}[d_{\mathcal{M}}(\delta;\mathbf{U}) \neq d]. \tag{14}$$

The trace includes $D = d$. The same latent state is used in the factual and intervened worlds for each posterior draw. A point-identified state makes this probability zero or one, but does not remove uncertainty about the structural equations or their future versions.

A distinct optimisation target, for $0 \le \alpha < 1$, is the posterior chance-constrained margin

$$m_\alpha(\tau) = \inf\{\mathrm{cost}(\delta) : \delta \in \Delta, \rho_{\mathcal{M}}(\delta \mid \tau) \ge 1-\alpha\}. \tag{15}$$

The intervention is chosen once, from the observed trace, and cannot vary with an unobserved posterior draw, so this quantity differs from an average of state-specific oracle minima. We assume that the intervention catalogue and the events involved have the measurability these probabilities require.

Posterior uncertainty about $\mathbf{U}$, uncertainty across compatible mechanisms and variation in future deployment are different quantities. For a nonempty declared model class $\mathfrak{M}$, one may report $\inf_{M\in\mathfrak{M}} \rho_M$ and $\sup_{M\in\mathfrak{M}} \rho_M$, and an empty compatible class signals a model failure. Repeated fresh service calls estimate fresh-run success, which is not the same as the counterfactual probability of the historical interaction, and $\rho_{\mathcal{M}}$ is not a measure of model-version drift.

A Monte Carlo estimate of ρ needs a stated posterior sampler and a statement of its sampling uncertainty, and probability calibration is assessed on held-out units with simulator ground truth. For stochastic validation, the candidate must be frozen before independent validation draws are used, because fixed-sample intervals do not cover repeated selection or optional stopping. Several valid or ε-optimal candidates may be reported with their costs, without any implication that the explanation is unique or that a single change explains the event. For the finite veto class below, a uniform bound gives an alternative that remains valid when search and estimation use the same posterior samples (Theorem 6.6).

6 Posterior Intervention Search

The distinction between a known latent state and an unresolved historical state changes both the interpretation and the complexity of an audit. Throughout this section we fix a layered audit problem (Definition 3.4). Write $\rho(\delta \mid \tau)$ for (14) and define $m_{\alpha,i}(\tau)$ by restricting (15) to the feasible pure interventions Δ_i. All empty feasible-target sets have margin $+\infty$.

6.1 Pure layers under unresolved latent state

The successful component in Proposition 3.10 can depend on the latent state. Selecting one intervention before that state is resolved reverses the order of choices: the known-state analysis fixes a state and then picks a cheapest action, whereas the auditor must pick one action that works across every likely state.

Proposition 6.1. *For every $\alpha \in [0, 1/2)$ and every $K > 0$ there is a layered audit problem in which layer-local pivotality holds at every posterior-supported state, so that $m(\mathbf{u}) = \min_i m_i(\mathbf{u})$ for all such states, and yet $m_\alpha(\tau) < \infty$ and*

$$\min_i m_{\alpha,i}(\tau) \ge K\, m_\alpha(\tau).$$

Proof. Let root inputs $X = Z = 1$, a clause variable $C = 1$, and an unobserved fair bit U determine

$$D = Z \wedge \begin{cases} X, & U = 0, \\ C, & U = 1. \end{cases}$$

The factual trace $X = Z = C = D = 1$ leaves U uniform. Permit $X := 0$, $C := 0$ and $Z := 0$ at costs $1, 1, M$, respectively, where $M > 2$. Their unions are feasible and their costs add. Variables X, Z are in the input layer and C in the second (clause) layer. At $U = 0$, $X := 0$ suffices as intervention. At $U = 1$, $C := 0$ suffices. Every successful intervention contains either the appropriate unit cost action or $Z := 0$, so a feasible pure component succeeds and the statewise optimum is one. Each cheap action alone has posterior success $1/2$, whereas their mixed union succeeds surely at cost two. A pure intervention succeeding with probability above $1/2$ must contain $Z := 0$. Thus, for $0 \le \alpha < 1/2$,

$$m_\alpha(\tau) = 2, \qquad \min_i m_{\alpha,i}(\tau) = M.$$

Taking $M \ge 2K$ proves the claim. Excluding $Z := 0$ instead makes the pure posterior margin infinite. □

For example, at $M = 100$ and requested success $Q = 0.9$, the input-only audit costs 100, the joint audit costs 2, and a state-aware oracle costs 1. Averaging the oracle costs would still give 1, but would not produce an implementable common action. At $Q \le 1/2$ either cheap action suffices; the gap therefore reflects a confidence requirement as well as uncertainty. Forcing a prerequisite to fail identifies which evidence or verdict the recorded approval depended on.

Revealing U would allow a cost-one intervention, so the gap is caused by uncertainty alone: every known-state instance is solved by a pure intervention. More information does not always help, however; on some traces it reveals a harder or infeasible case.

Proposition 6.2. *In a layered audit problem with $L \ge 1$ layers, suppose layer-local pivotality holds at almost every $\mathbf{u}$ conditional on τ. Then for any $Q \in (0, 1]$,*

$$\min_i m_{1-Q/L,i}(\tau) \le m_{1-Q}(\tau). \tag{16}$$

The factor L cannot be improved in general.

Proof. For any δ with $\rho(\delta \mid \tau) \geq Q$, let $J(\delta) = \{i : \delta_i \in \Delta_i\}$. Statewise pivotality and the union bound give

$$Q \leq \Pr\left(\bigcup_{i \in J(\delta)} \{D_{\delta_i} \neq d\} \,\middle|\, \tau \right) \leq \sum_{i \in J(\delta)} \rho(\delta_i \mid \tau).$$

Some feasible component therefore succeeds with probability at least Q/L, and its cost is at most $\mathrm{cost}(\delta)$. Taking infima proves (16), using approximating sequences when the infimum is not attained. An infinite right-hand side is immediate. For sharpness, take L latent states with equal probability, constant-one variables $A_1, \ldots, A_L$ in distinct layers, and $D = A_U$. Permit only their zero overrides and unions. Every successful intervention at a fixed state has a successful pure component. The union of all L actions succeeds surely, while each nonempty pure intervention succeeds with probability $1/L$. No pure intervention exceeds that probability, so no factor smaller than L can replace L in (16). □

6.2 A veto class with posterior coverage

We now give structural conditions under which posterior search has a useful diminishing-returns property, known as submodularity: adding an action to a larger set never helps more than adding it to a smaller one. Consider a finite acyclic Boolean SCM whose equations have the form

$$X_v = b_v(\mathbf{U}) \wedge \bigwedge_{w \in K_v(\mathbf{U})} X_w, \qquad K_v(\mathbf{U}) \subseteq \mathrm{pa}(v), \tag{17}$$

where $b_v \in \{0, 1\}$ and an empty conjunction is one. The joint latent state is allowed to couple all mechanisms. The active parent set K_v depends only on $\mathbf{U}$, not on endogenous values changed by an intervention. These equations describe veto or prerequisite checks, not arbitrary LLM mechanisms or adaptive protocol selection.

Condition on a supported factual trace containing $D = 1$. Let I be a finite whitelist of nonterminal variables, and let δ_S force $X_v := 0$ for every $v \in S \subseteq I$. Every subset is admissible and

$$c(S) := \mathrm{cost}(\delta_S) = \sum_{v \in S} c_v, \qquad c_v > 0.$$

Such actions are controlled verdict failures or input withdrawals. Their semantics do not encompass arbitrary numerical or textual changes. For fixed $\mathbf{u}$, retain the edge $w \to v$ exactly when $w \in K_v(\mathbf{u})$, and let $R(\mathbf{u})$ be the ancestors of D in this active graph that belong to I.

Theorem 6.3 (Posterior coverage). *Under the assumptions of this subsection, for every $S \subseteq I$ and every latent state $\mathbf{u}$ compatible with τ,*

$$D_{\delta_S}(\mathbf{u}) = 0 \quad \Longleftrightarrow \quad S \cap R(\mathbf{u}) \neq \varnothing, \tag{18}$$

and consequently

$$F_\tau(S) := \rho(\delta_S \mid \tau) = \mathbb{E}[\mathbf{1}\{S \cap R(\mathbf{U}) \neq \varnothing\} \mid \tau]. \tag{19}$$

The set function F_τ satisfies $F_\tau(\varnothing) = 0$ and is monotone and submodular.

Proof. Factual $D = 1$ forces every active ancestor of D, and its base value b_v, to be one. Forcing an active ancestor to zero propagates zero along an active path to D. Additional overrides on the path also force zero and cannot restore approval. If no override is an active ancestor, the mechanisms and inputs on the active ancestry remain factual, so $D = 1$. This proves (18). Posterior averaging gives (19).

For a fixed set R, adding v to S increases $\mathbf{1}\{S \cap R \neq \varnothing\}$ by

$$\mathbf{1}\{v \in R\}\, \mathbf{1}\{S \cap R = \varnothing\}.$$

This is nonnegative and decreases as S grows. Averaging under the fixed posterior proves monotonicity and the diminishing-returns inequality

$$F_\tau(S \cup \{v\}) - F_\tau(S) \geq F_\tau(T \cup \{v\}) - F_\tau(T)$$

for $S \subseteq T$ and $v \notin T$, which is submodularity. Normalisation follows from $F_\tau(\varnothing) = 0$. □

The proof is a coverage argument of the kind used in influence maximisation [26]. It holds even when conditioning on the trace correlates the active edges, because the posterior stays fixed while S is chosen: interventions do not change the historical conditioning event. The property is ordinary submodularity for a simultaneous intervention; a policy that gathers observations between actions would need its adaptive variant. Success probabilities strictly between zero and one arise only when $R(\mathbf{U})$ varies with the latent state.

6.3 Hardness and a greedy cost–confidence guarantee

For $Q \in (0, 1]$, write

$$C^*_Q = \min\{c(S) : S \subseteq I,\ F_\tau(S) \geq Q\}, \tag{20}$$

with value $+\infty$ if infeasible. Note that this equals $m_{1-Q}(\tau)$ for the restricted veto catalogue and that finiteness of I guarantees attainment when the target is feasible.

Theorem 6.4 (Hardness of posterior intervention). *Computing C^*_1 is NP-hard, even when the SCM has only constant roots and a single terminal veto mechanism, the posterior is uniform on a finite set, and the action costs are positive integers. The decision problem of whether $C^*_1 \leq B$ for a given budget B is NP-complete when the active-parent incidence table and the posterior are given explicitly.*

Proof. Take a weighted Set Cover instance with nonempty universe J, sets $A_1, \ldots, A_n$ whose union is J, and positive integer costs c_i. Introduce roots $X_i = 1$, let U be uniform on J, and set

$$D = \bigwedge_{i : U \in A_i} X_i.$$

The trace $X_1 = \cdots = X_n = D = 1$ leaves U uniform. Action i forces $X_i := 0$ at cost c_i. A selected set S flips the decision at $U = j$ exactly when $j \in \bigcup_{i \in S} A_i$. Thus $F_\tau(S) = 1$ precisely when the chosen sets cover J, and C^*_1 equals the minimum cover cost. Set Cover is NP-complete even with unit costs [25]. The reduction has polynomial size. A proposed S can be checked against the supplied incidence table and budget in polynomial time, giving membership in NP. At a specified j, choosing a cheapest i with $j \in A_i$ is optimal. □

For any fixed rational target $Q_0 \in (0, 1)$, add posterior mass $1 - Q_0$ on a state with no actionable ancestors and distribute mass Q_0 uniformly over the cover states. Achieving Q_0 again requires a complete cover, so exact optimisation remains hard. In this construction, Q_0 is the maximum attainable success probability. The

difficulty concerns choosing one common intervention. In satisfiability terms, each posterior state contributes one clause of positive literals, so posterior search in the veto class has a cost-constrained weighted partial-MaxSAT encoding. For succinct Boolean mechanisms with polynomial-time evaluation and independent fair latent bits, selecting an intervention before the random state contains the E-MAJSAT problem, which is $\mathrm{NP}^{\mathrm{PP}}$-complete [27]. This complexity comparison concerns that finite Boolean encoding, rather than unrestricted SCMs. Appendix F develops the connection.

An approximation follows from standard submodular-cover analysis [50]. Cap the objective at the requested confidence: $f_Q(S) = \min\{F_\tau(S), Q\}$. Starting from $S = \varnothing$, repeatedly add an unused action maximising

$$\frac{f_Q(S \cup \{v\}) - f_Q(S)}{c_v}, \tag{21}$$

and stop as soon as $f_Q(S) \geq Q - \varepsilon$.

Theorem 6.5 (Greedy posterior audit bound). *Let $C^*_Q < \infty$ and $0 < \varepsilon < Q$, and suppose F_τ is evaluated exactly. The greedy rule (21) terminates after at most $|I|$ additions and returns a set S_{gr} with*

$$F_\tau(S_{\mathrm{gr}}) \geq Q - \varepsilon, \tag{22}$$

$$c(S_{\mathrm{gr}}) \leq \left(1 + \log \frac{Q}{\varepsilon}\right) C^*_Q. \tag{23}$$

Proof. Truncation preserves monotonicity and submodularity. Let O achieve C^*_Q. At a current set S, put $r = Q - f_Q(S)$. Before stopping, $r > \varepsilon$, and

$$r = f_Q(S \cup O) - f_Q(S) \leq \sum_{v \in O \setminus S} [f_Q(S \cup \{v\}) - f_Q(S)].$$

Some action therefore has gain per cost at least r/C^*_Q, and the greedy choice does too. If its gain is h, its cost is at most $C^*_Q h/r \leq C^*_Q$ because the objective is capped. The positive gain also ensures termination. For all but the last addition, the new residual is still greater than ε, and

$$\frac{h}{r} \leq \log \frac{r}{r-h}.$$

Summing telescopes to at most $\log(Q/\varepsilon)$. The last addition costs at most C^*_Q, proving (23). The stopping rule gives (22). □

This bound allows confidence slack: cost is compared with an optimum at Q, while the output reaches $Q - \varepsilon$. It does not assert the same approximation factor at an exactly attained target. For an explicit posterior on k states, exact evaluation is cheap: the k ancestor sets are computed once and reused for every coverage query, without enumerating the $2^{|I|}$ interventions. A general posterior may require expensive inference, which Section 6.4 replaces by sampling.

6.4 Guarantees from finite posterior samples

Let $n = |I|$ and let $\mathbf{U}_1, \ldots, \mathbf{U}_N$ be independent draws from the declared posterior. Define

$$\widehat{F}_N(S) = \frac{1}{N} \sum_{j=1}^{N} \mathbf{1}\{S \cap R(\mathbf{U}_j) \neq \varnothing\}.$$

The empirical function is itself monotone submodular.

Theorem 6.6 (Uniform posterior estimation). *For $\beta \in (0,1)$ and $\eta > 0$, if*

$$N \geq \frac{n \log 2 + \log(2/\beta)}{2\eta^2}, \tag{24}$$

then, with probability at least $1 - \beta$ over the draws,

$$\sup_{S \subseteq I} |\widehat{F}_N(S) - F_\tau(S)| \leq \eta. \tag{25}$$

In particular, the bound holds for a set selected using those same samples.

Proof. For each fixed S, Hoeffding's inequality [20] bounds the failure probability by $2\exp(-2N\eta^2)$. A union bound over the 2^n subsets gives $2^{n+1}\exp(-2N\eta^2) \leq \beta$ under (24). On the resulting event, the inequality holds simultaneously for all sets, including a data-selected one. □

Corollary 6.7. *Let $Q, \eta, \varepsilon > 0$ with $Q + 2\eta + \varepsilon \leq 1$, suppose $C^*_{Q+2\eta+\varepsilon} < \infty$, and let N satisfy (24). Let S_{gr} be the output of the greedy rule applied to $\widehat{F}_N$ with cap $T = Q + \eta + \varepsilon$ and stopping threshold $T - \varepsilon = Q + \eta$. Then, with probability at least $1 - \beta$, the rule reaches its threshold and*

$$F_\tau(S_{\mathrm{gr}}) \geq Q, \tag{26}$$

$$c(S_{\mathrm{gr}}) \leq \left(1 + \log \frac{T}{\varepsilon}\right) C^*_{Q+2\eta+\varepsilon}. \tag{27}$$

Proof. On (25), a true optimum at $Q + 2\eta + \varepsilon$ has empirical success at least T. Apply Theorem 6.5 to the empirical objective and compare its T-optimum with this feasible set. The selected intervention has empirical success at least $Q + \eta$ and hence true success at least Q. □

If the empirical target is unattainable, the search abstains instead of reporting success. The cost bound (27) compares with the optimum at the higher confidence $Q + 2\eta + \varepsilon$ because the optimal cost can jump as the confidence rises. At $Q = 1$ this sampled guarantee is unavailable: finite sampling alone cannot generally certify zero posterior failure probability. If the sampling law is within total variation distance ζ of the declared posterior, every event probability moves by at most ζ; replacing η by $\eta + \zeta$ in the confidence margins, but not in (24), extends the corollary.

6.5 Limits of the coverage guarantee

The coverage guarantee has three boundaries. The first is complementarity: for $D = X_1 \wedge X_2$ with factual $X_1 = X_2 = 0$ and actions $X_i := 1$, neither action succeeds alone but their union does, so the success function is not submodular. General threshold rules, value-dependent protocol switching and strategic responses can introduce the same effect.

The second is model uncertainty. The coverage functions $F_1(S) = \mathbf{1}\{a \in S\}$ and $F_2(S) = \mathbf{1}\{b \in S\}$ have a non-submodular lower envelope, since $\min(F_1, F_2)$ requires both actions. A robust lower probability over a model class (Sections 5 and 7) therefore need not be submodular, even when every model's coverage function is.

The third is feasibility and cost. Coupled feasibility or nonadditive costs break steps of the greedy proof. A predefined bundle of interventions can act as a single action when every union remains

feasible, the coverage representation holds and costs stay additive; the layer structure alone guarantees none of these.

7 Identification and Partial Identification

Evaluating a known SCM and identifying a counterfactual from observed laws are different tasks. A fitted mechanism can predict observed decisions accurately while assigning the wrong coupling between factual and counterfactual outcomes. Identification results require a specified model class and causal structure [32], and a discrete Gumbel-max SCM is one explicit coupling choice [33]. We record a positive scalar result and two limitations.

Proposition 7.1. *Let $Z = f_A(U)$, where A denotes the observed parents of Z, $U \sim \mathrm{Unif}(0,1)$ is independent of A, and each f_a is continuous and strictly increasing. Suppose the conditional distribution functions $F_a(z) = \Pr(Z \le z \mid A = a)$ are identified, continuous and strictly increasing on the support of Z given $A = a$. Then, within this model class, for almost every factual observation $(A = a, Z = z)$ and every a' at which $F_{a'}$ is identified,*

$$U = F_a(z) \qquad \text{and} \qquad Z_{a'} = F_{a'}^{-1}(F_a(z)),$$

so the counterfactual $Z_{a'}$ is identified.

The derivation is given in Appendix C.

For continuous parents, conditional laws are determined only almost everywhere, so querying exceptional or unsupported values needs additional regularity, and estimating the conditional CDFs adds statistical error. Observing only $D = \mathbf{1}[Z \ge \theta]$ generally reveals a rank interval rather than an individual's rank. A binary decision-boundary sweep therefore neither reveals a unique latent score nor verifies monotonicity in the exogenous noise, which is a different assumption from monotonicity in an observed financial input.

Proposition 7.2. *Let D_0, D_1 be binary potential outcomes. Assume treatment A is independent of this pair, $D = D_A$, and both treatment arms have positive probability. If the experimental marginals $p_a = \Pr(D_a = 1)$ are known and $p_0 < 1$, then*

$$\frac{\max(0, p_1 - p_0)}{1 - p_0} \le \Pr(D_1 = 1 \mid A = 0, D = 0) \le \min\left(1, \frac{p_1}{1 - p_0}\right). \tag{28}$$

Both bounds are sharp in the absence of further restrictions on the coupling.

The derivation is given in Appendix C.

These are bounds for the stated observation, in the tradition of probabilities of causation [43]. Additional trace information requires conditioning the compatible response-type models on that information, and estimated marginals carry sampling uncertainty. For independent fair bits A, U, the models $D = U$ and $D = A \mathrm{XOR} U$ have the same observational and experimental marginals, $p_0 = p_1 = 1/2$. After observing $A = 0, D = 0$, their flip probabilities under $A := 1$ are respectively zero and one. No estimator of these marginals can distinguish the two couplings.

General vector monotonicity is insufficient to identify a coupling. Appendix C gives the rotation counterexample and explains why imposing a transport family adds assumptions beyond monotonicity.

Epistemic labels. An audit reports *exact under a declared replay model*, *identified within a stated model class*, *partially identified* (with bounds), or *surrogate-dependent*. Known downstream equations propagate uncertainty in their inputs rather than removing it. A label applies to the complete query, names the conditioning information, support and coupling assumptions, and is accompanied by the statistical uncertainty.

8 Counterfactual Analysis Under Strategic Behaviour

A disclosure mechanism maps a counterparty's private state and information to reported evidence. Writing it as a function says nothing about why the counterparty reports what it does, so we specify a finite committed-policy model that gives behavioural counterfactuals an explicit response concept.

Timing, information and utility. The lender publicly commits to policy π. The counterparty observes its type ϑ and π, and chooses a report z from a nonempty finite catalogue $Z(\vartheta)$. The lender then follows its announced verification and decision mechanisms. Write $Q(\pi, z, \vartheta, U_D)$ for the terminal outcome, allowing verification to reveal some true-state information, and $v(\vartheta, z; \pi)$ for the counterparty's expected utility. This utility includes the approval benefit, disclosure effort and expected verification penalties, with the expectation taken over the declared subsequent noise.

For $\epsilon \ge 0$, define

$$\mathrm{BR}_\epsilon(\vartheta; \pi) = \Big\{ z \in Z(\vartheta) : v(\vartheta, z; \pi) \ge \max_{z' \in Z(\vartheta)} v(\vartheta, z'; \pi) - \epsilon \Big\}. \tag{29}$$

This set is nonempty by finiteness. A declared tie-breaking rule supplies the structural equation

$$z = S\big(\mathrm{BR}_\epsilon(\vartheta; \pi), U_R\big), \qquad \mathbf{E} = R(\vartheta, z), \tag{30}$$

where R compiles the report into an evidence package. The report z is an endogenous variable, and replacing S or R is a mechanism intervention. The model describes a response to a committed lender and makes no claim about a mutual-adaptation equilibrium. For an observed or learned policy, regret against the complete finite report catalogue tests the best-response assumption, whereas an incomplete deviation search gives only a lower bound on regret.

Outcome stability under approximate responses. Small utility regret need not preserve a counterfactual decision. Proposition D.1 in Appendix D gives an explicit condition: if utility estimates have uniform error at most η, an ϵ-best response has true regret at most $\epsilon + 2\eta$. It agrees with a true best response on the decision when the utility gap to every report with a different outcome exceeds that bound. The appendix gives the proof, a sharpness example and a population bound. These guarantees assume the outcome map is correct.

Different intervention targets. A true-state intervention changes ϑ and regenerates disclosure and downstream state. A disclosure override fixes $z := z'$ while retaining ϑ, and changing the reporting rule replaces its mechanism. A policy intervention $\pi := \pi'$ can either hold observed disclosure fixed or regenerate the response through (29). These yield different counterfactuals. If response selection is unspecified, report the range of outcomes or outcome

probabilities over $\mathrm{BR}_\epsilon(\vartheta; \pi')$ rather than selecting a favourable response. Changing a policy parameter also recomputes every clause that uses it, whereas overriding one clause verdict changes only that node's mechanism for the current evaluation.

Sequential extension. For a finite interaction, unroll the graph in time: action z_t depends only on the acting agent's available history H_t and its private state, and H_{t+1} is generated from H_t, the actions and current noise. An intervention at time t retains the factual prefix and regenerates the dependent suffix, using the same mechanism/time-indexed noise in both worlds. Fixing the complete recorded transcript instead defines a controlled decision-stage audit, as in Section 3.2, which does not measure the total effect of an earlier disclosure. The experiment below implements a report, a possible document request, a verifiable reply and a terminal lender decision, with exact finite report best responses.

Truthfulness and authorisation. For a policy-update outcome J, the predicate $\mathbf{1}[J_{\text{truthful}}(\mathbf{u}) = J_{\text{factual}}(\mathbf{u})]$ measures *truthfulness invariance*, provided truthful disclosure is specified and feasible. Its complement is disclosure dependence. An independent authorisation predicate records whether the update was permitted. Neither predicate implies the other: a permitted privacy choice can affect an authorised update, while a prohibited but causally redundant action can leave the same update under truthful disclosure. Unknown true state or response mechanisms require posterior uncertainty or bounds.

9 Experiments

We examine three choices an auditor must make: whether to search across decision stages, whether to regenerate downstream state, and whether to allow the counterparty to respond. Each can change the answer even when the recorded interaction is reproduced perfectly. Our synthetic models provide exact counterfactual outcomes and minimum costs, so every discrepancy is measured against a known answer. We compare search methods first with the latent state known and then under posterior uncertainty, before examining propagation and strategic response.

Evaluation design. The offline suite contains 2,800 finite instances: seven regimes, each with 20 seed blocks and 20 instances per block. Every method shares the intervention catalogue, synthetic assignment costs, feasibility checker and SCM evaluator. Exhaustive enumeration supplies a global reference optimum. We distinguish *valid returns*, which actually change the outcome, from *optimal returns*, which also attain the global minimum cost. Both rates include every instance, including abstentions and infeasible targets. The primary budget is 16 candidate evaluations plus at most one validation. Budgets 4 and 64 provide sensitivity checks.

9.1 A Valid Change Can Still Be Needlessly Expensive

Joint search evaluates feasible candidates in increasing cost order. Single-layer search does the same within the pure catalogue. Random search returns the cheapest success among the feasible candidates it samples. Table 2 reports the three regimes that separate the methods. A method can return valid changes without returning cheap ones.

Table 2: Known-state search: valid and optimal returns (%) at a budget of 16 evaluations, 400 instances per regime. Valid: the returned intervention changes the decision. Optimal: it also attains the exhaustive joint optimum. The four control regimes are summarised in the text.

	Joint, ordered		Single layer		Joint, random	
Regime	Valid	Opt.	Valid	Opt.	Valid	Opt.
Mixed-layer	100.0	100.0	100.0	0.0	100.0	88.0
Protocol	89.8	89.8	100.0	42.8	100.0	15.3
Nonmonotone	100.0	100.0	0.0	0.0	100.0	100.0

In the mixed-layer repair regime, *single-layer search succeeds in all* 400 *instances but finds the global optimum in none.* Joint search finds the optimum in all 400. The mechanism is the example in Remark 3.12: a root-level evidence change and a clause override together cost 2, whereas repairing both clauses costs $M + 1$. A deterministic sweep of M from 2 to 1,024 yields pure/joint cost ratios from 1.5 to 512.5, the ratio $(M+1)/2$ of the construction. Four control regimes behave as designed. In the veto regime, both ordered searches find every optimum, as layer-local pivotality predicts; with coupled feasibility and with zero-cost ties, all three methods do; and for infeasible targets, no method returns a change.

Protocol interactions expose a budget tradeoff. At 16 evaluations, joint search finds a valid global optimum in 89.75% of instances. Single-layer and random search each return valid changes in 100%, but global optima in only 42.75% and 15.25%, respectively. The paired validity difference between joint and single-layer search is -10.25 percentage points, with 95% seed-block bootstrap interval $[-13.5, -7.0]$. At budget 64, joint search reaches 100% optimal returns, while the single-layer rate remains 42.75%; random search reaches 64.5%. More evaluations resolve a search-budget shortfall, but cannot recover solutions excluded by the intervention family.

9.2 Searching Under Posterior Uncertainty

Proposition 6.1 shows that single-layer search can fail under uncertainty even when it is exact at every known state. To measure how often this happens, we draw 1,000 random veto SCMs of the form (17) in 20 seed blocks. Each has seven intervenable variables in three layers, a terminal approval, and 24 latent states with random active parents and base values. The trace records two variables and the approval, leaving a posterior over a median of 9 states; action costs are random integers from 1 to 9. At each of the 7,153 known states admitting a successful intervention, single-layer search is exact. States with no actionable ancestor are infeasible for both joint and pure search. Under the posterior it is not (Table 3): among the 765 instances feasible at confidence 0.8, the best single-layer intervention is infeasible or more expensive than the joint optimum in 291 (38.0%). At $Q = 1$, the corresponding count is 168 of 374 feasible instances (44.9%). Greedy search stopped at Q costs 8–13% more than the optimum on average and finds it in 60–75% of instances. With confidence slack 0.05, the bound of Theorem 6.5 held in all 2,614 cases.

Further finite checks confirm the layer-gap examples, the active-ancestor identity, the coverage properties and the Set Cover encoding, over 45,312 intervention/state replays. With an analytic

Table 3: Search under posterior uncertainty in 1,000 random veto models. n: instances in which some intervention reaches confidence Q. Single layer worse: the best single-layer intervention is infeasible or costs more than the joint optimum. Greedy: cost of greedy search stopped at Q, relative to the joint optimum. Brackets: 95% seed-block bootstrap intervals.

Q	n	Single layer worse (%)	Greedy cost ratio	Greedy optimal (%)
0.5	948	9.3 [7.4, 11.4]	1.11 [1.10, 1.12]	74.9
0.8	765	38.0 [35.0, 41.1]	1.13 [1.12, 1.15]	64.3
0.9	527	42.9 [39.5, 46.6]	1.13 [1.12, 1.14]	60.5
1.0	374	44.9 [40.7, 49.0]	1.08 [1.07, 1.09]	68.7

posterior on 2,000 held-out units, success probabilities estimated from 256 posterior draws per query have mean absolute error 0.0219 (seed-block interval $[0.0212, 0.0227]$).

9.3 Freezing State Can Invent or Remove Explanations

The correct evaluator regenerates all non-intervened descendants. A frozen-clause ablation instead inserts the proposed values into the factual state and re-evaluates only the terminal decision. In all 400 nonmonotone instances, it claims a decision change that disappears under correct replay. Here $C = 1 - X$ and $D = X \wedge C$, with factual $X = 0$: setting $X := 1$ appears to approve if C is incorrectly left at 1, but actually changes C to 0. A valid mixed intervention must explicitly override C as well. This is why factual-value internal overrides remain in the catalogue.

A separate ablation prunes every variable that is not an ancestor of the outcome. It loses all 400 solutions in the coupled-feasibility regime: changing the decision-relevant accounting field requires an accompanying certificate, even though the certificate has no direct causal path to the decision. The shared feasibility checker rejects the incomplete edit. Together, the two controls isolate two requirements of a correct audit: propagate consequences, and preserve the constraints that make a change admissible.

9.4 A Policy Change Also Changes the Borrower's Behaviour

The sequential experiment gives the borrower private income band $\vartheta \in \{0, 1, 2\}$ and a choice of reported band z. The lender may request a verifiable document before deciding. We enumerate the borrower's best responses under a committed policy, with approval benefit 5, effort $k|z - \vartheta|$, verification probability q and penalty F for a detected mismatch. The design crosses $q \in \{0, 1/4, 1/2, 1\}$, $F \in \{0, 2, 8\}$, $k \in \{0, 1/2, 2\}$, three types and 20 noise seeds, giving 2,160 combinations. Each mechanism draws its own randomness, which is held fixed across the compared worlds, and all tied best responses are retained.

Consider a borrower with true band 1, verification probability zero and effort weight $1/2$. Under eligibility threshold 1, the borrower truthfully reports 1 and is approved. Raising the threshold to 2 rejects the unchanged report. But the borrower can respond by reporting 2: utility falls only from 5 to 4.5, so this is the new best response and approval persists. The static audit predicts that tightening the rule changes the decision; the behavioural audit predicts continued approval. This example is one of the archived traces.

Across the complete design, these two policy queries disagree in 607/2,160 combinations (28.1%; seed-block interval $[27.0, 29.3]$%).

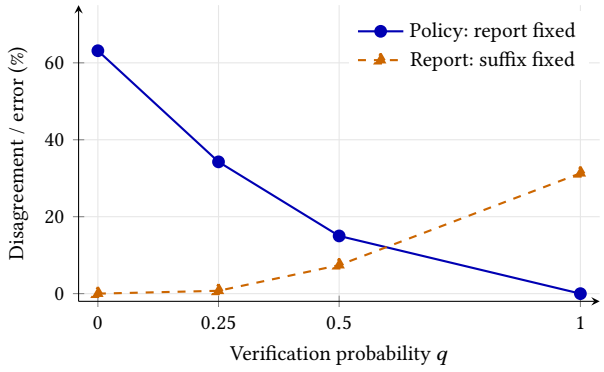


Figure 3: Different audit shortcuts fail in different verification regimes. Solid blue: disagreement with the behavioural policy query when the report is held fixed (540 combinations per q). Dashed orange: wrong report-intervention decisions when the factual suffix is reused (1,080 interventions per q). Points summarise the complete finite design; lines guide the eye.

For a separate query that changes the report while keeping policy fixed, reusing the factual document instead of regenerating the request and reply gives 426/4,320 wrong decisions (9.9%; interval $[9.4, 10.3]$%), with two alternative reports per interaction.

Figure 3 shows how the discrepancies depend on verification. Policy-query disagreement falls from 63.1% without verification to zero with certain verification. Over the same settings, frozen-suffix error rises from zero to 31.3%: changing a previously ineligible report can now trigger a document request absent from the factual transcript. More verification makes that missing downstream response consequential, so the two shortcuts fail in opposite regimes.

9.5 Affine and Identification Controls

For 4,000 affine instances in dimensions 2, 5, 10, 20, we vary bounded, immutable, one-sided and blocked coordinates, including observations exactly on the threshold. Exact rational calculations find 1,145 infeasible targets. Another 1,750 cases have finite margins that are not attained. Every feasible case returns a checked successful witness within additive cost 0.01 of the infimum. Independent small-box vertex checks and the unrestricted formula test the reference implementation. These numerical checks do not replace the proofs.

Appendix E.3 describes the scalar rank, binary bounds and posterior calibration controls. They separate estimation error from wrong coupling or mechanism assumptions.

Scope. The known-state regimes are small synthetic models, each built to isolate one mechanism: they show what an audit choice does, not how often it matters in deployed systems. The random veto models reuse the generator settings of our theorem checks rather than a calibrated lending model, and greedy search without confidence slack lies outside the guarantee of Theorem 6.5. Catalogues are enumerated, so comparisons count outcome queries; construction and sorting costs are recorded separately. The theorem checks confirm that the implementation matches the proofs and are not a scalability study. Borrower behaviour is an exact best response over three reports and says nothing about the rationality of language-model agents. Bootstrap intervals resample whole seed blocks 2,000 times and are descriptive, without multiplicity adjustment. The study is offline, with no private data or model calls; the local reproduction package contains the complete records and code, with the protocol and further controls in Appendix E. Scalability, the fidelity of learned mechanisms and usefulness to human principals remain open empirical questions.

10 Conclusion

Counterfactual audit can be formulated as constrained intervention search with explicit propagation, cost and epistemic assumptions. Layer-restricted margins provide upper bounds on the joint margin. Pure exactness or approximation requires a successful feasible replacement property, and without it mixed interventions can be arbitrarily cheaper. Exact affine references, finite enumeration and bounded-width optimisation distinguish a decision-boundary distance, a successful witness and an optimisation certificate.

Uncertainty about the historical latent state creates an additional gap: pure exactness at every state need not survive posterior optimisation at fixed confidence. Statewise pivotality instead supplies a sharp confidence relaxation across layers. For the specified Boolean veto class, posterior success is a coverage function, giving an NP-hardness result together with greedy cost–confidence and finite-sample guarantees. These conclusions depend on the intervention direction, fixed active-dependency semantics and the feasibility and cost assumptions, and do not extend to arbitrary agent mechanisms or to worst-case model uncertainty.

The audit must also distinguish evaluation correctness from causal identification. Scalar rank preservation supports a positive identification result, while indistinguishable binary and vector mechanisms demonstrate limits of observational agreement and general monotonicity. Compositional replay checks certify evaluation in a declared SCM under trusted interfaces, but neither authenticate the underlying evidence nor establish optimality. A finite committed-policy disclosure model separates interventions on true state, reports and policies, including the effect of behavioural response. An outcome-gap condition states when utility-estimation error and approximate best responses preserve the audited outcome.

The experiments turn these distinctions into measurable quantities. In 1,000 random veto models, pure search is infeasible or more costly in 38.0% of the instances where a joint intervention reaches confidence 0.8, although pure search is exact at each feasible known state. The sequential models show why policy interventions must distinguish fixed disclosure from a regenerated response, and why early edits require suffix regeneration. All of these are reference results in synthetic models, not measurements of deployed negotiating agents.

An arena of heterogeneous language-model agents requires the same explicit catalogues, timing and response semantics. Applying the coverage theorem also requires the restricted veto mechanisms and intervention direction. Complementarities and mutual strategic adaptation need further theory. For hosted models, regenerated continuations generally estimate fresh-run effects rather than identified historical counterfactuals. Scaling search, validating fitted causal structures and assessing usefulness to human principals remain open. Appendix F sets out the associated satisfiability and random-threshold questions.

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A Layer Replacement and Search Certificates

This appendix supplies the derivations for Propositions 3.11 and 4.1. Partition assignments into layers and let m and m_i denote joint and pure-layer infima at a fixed compatible latent state, as in the main paper. If every successful δ has a feasible successful pure replacement γ with $\mathrm{cost}(\gamma) \le \kappa \mathrm{cost}(\delta)$, where $1 \le \kappa < \infty$, then

$$m \le \min_i m_i \le \kappa m.$$

The first inequality is feasible-set inclusion. For the second, each successful δ gives $\min_i m_i \le \mathrm{cost}(\gamma) \le \kappa \mathrm{cost}(\delta)$. Taking infima, or an approximating sequence if the infimum is not attained, proves the bound. Empty successful sets have infinite margins. Layer-local pivotality is the special case in which γ is a feasible successful component and monotone cost gives $\kappa = 1$.

Component feasibility matters separately from causal success. With factual $X = C = 0$ and $D = \mathbf{1}[X \ge 1 \text{ or } C = 1]$, allow only the empty intervention, $\{X := 1, C := 1\}$ of cost 2, and $\{X := 2\}$ of cost 3. Either component of the mixed action would change the outcome but is unavailable alone. The joint and pure margins are therefore 2 and 3.

Finite search. For a finite feasible catalogue evaluated exactly in nondecreasing cost order, the first success is minimal in that catalogue: every cheaper candidate has already failed. Before success, the smallest untested cost is a lower bound on the optimum. Only exhaustion proves infeasibility. Constructing and sorting the catalogue may take exponential work. Restricting to ancestors is sound only when deleting other assignments preserves feasibility and cannot increase cost. A non-ancestor certificate required by an accounting constraint violates that assumption.

B Affine Reference Margins

For the experimental controls we summarise the boundary and box calculations of Sections 3.1 and 4. The full duality and adversarial-robustness discussion remains in the main text. Let $D = \mathbf{1}[r \ge 0]$, with $r = \mathbf{a}^T\mathbf{x} + b$ and $\mathbf{a} \ne 0$. For positive weights w_i, use $\|\delta\|_{w,\infty} = \max_i w_i|\delta_i|$ and set $A = \sum_i |a_i|/w_i$. The unrestricted distance to the decision boundary is

$$m_H = \frac{|r|}{A}.$$

Indeed, any boundary-reaching change has $|r| = |\mathbf{a}^T\delta| \le A\|\delta\|_{w,\infty}$. Equality is attained by

$$(\delta_H)_i = -\operatorname{sign}(r)\operatorname{sign}(a_i) m_H / w_i$$

for $a_i \ne 0$, with zero in the other coordinates. For $r = 0$ the zero change attains the boundary distance. If $r < 0$, reaching the boundary changes the decision. If $r > 0$, rejection requires strict crossing, so m_H is the unattained flip infimum and $(1+\epsilon)\delta_H$ succeeds for every $\epsilon > 0$. At $r = 0$ arbitrarily small changes along $-\mathbf{a}$ reject, but zero itself does not. Boundary distance, a successful witness and a minimum are therefore three distinct quantities.

Suppose $\ell_i \le \delta_i \le u_i$ with $\ell_i \le 0 \le u_i$. For $r < 0$, the maximal score improvement at cost at most t is

$$H(t) = \sum_{a_i > 0} a_i \min(u_i, t/w_i) + \sum_{a_i < 0} a_i \max(\ell_i, -t/w_i).$$

The exact box margin is $\inf\{t \ge 0 : H(t) \ge -r\}$. Coordinatewise maximisation proves the expression, and H is continuous, nondecreasing and piecewise affine. For $r \ge 0$, reverse the score direction and require improvement strictly greater than r. If the maximum only touches the threshold, rejection is infeasible. These formulas give exact reference margins for affine models, which the experiments use as controls.

C Identification Details

The first two derivations prove Propositions 7.1 and 7.2.

C.1 Scalar rank preservation

Suppose $Z = f_A(U)$, where $U \sim \mathrm{Unif}(0,1)$ is independent of observed parents A, and each f_a is continuous and strictly increasing. Assume the conditional CDFs F_a are identified, continuous and strictly increasing on their supports. Then, for almost every supported factual (a, z) and a supported target a', the rank and counterfactual are

$$U = F_a(z), \qquad Z_{a'} = F_{a'}^{-1}(F_a(z)).$$

Strict monotonicity and independence give $F_a(f_a(u)) = u$, and evaluating the target inverse at this rank proves the claim. The identification holds within the stipulated rank-preserving class. A thresholded observation identifies only an interval of ranks. Conditional laws at exceptional or unsupported parent values require additional assumptions, and fitted CDFs add sampling error.

C.2 Sharp binary interval

Let A be independent of binary potential outcomes (D_0, D_1), with $D = D_A$ and both arms represented. Put $p_a = \Pr(D_a = 1)$ and $p_{11} = \Pr(D_0 = 1, D_1 = 1)$. Nonnegative cell probabilities imply

$$\max(0, p_0 + p_1 - 1) \le p_{11} \le \min(p_0, p_1).$$

For $p_0 < 1$, the query after $A = 0, D = 0$ is $(p_1 - p_{11})/(1 - p_0)$. Substituting the two endpoints yields (28), and the endpoint tables attain it. These are the standard response-type bounds for the specified conditioning event [43]. Additional evidence requires conditioning the compatible response types on that evidence.

Limits of vector monotonicity and transport. Let U be uniform on the unit disk, independent of binary A, and let R_θ be planar rotation, $0 < \theta < \pi/2$. Compare $f_0(u) = u$, $f_1(u) = R_\theta u$ with $\widetilde{f_0}(u) = u$, $\widetilde{f_1}(u) = R_{-\theta}u$. Every mechanism is bijective and strongly operator-monotone because $v^T R_{\pm\theta} v = \cos(\theta)\|v\|_2^2 > 0$ for $v \ne 0$. Rotation invariance gives identical conditional outcome laws, yet $R_\theta z \ne R_{-\theta} z$ for $z \ne 0$. Thus general operator monotonicity does not identify vector counterfactuals.

Fixing an absolutely continuous latent law and requiring Brenier mechanisms for quadratic cost imposes stronger assumptions than operator monotonicity. Transport-based constructions [9] must be interpreted relative to that chosen family. Even the counterfactual composition of two linear Brenier mechanisms need not be monotone:

$$A = \begin{pmatrix} 10 & 0 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}, \quad BA^{-1} = \begin{pmatrix} 0.1 & 2 \\ 0.2 & 5 \end{pmatrix}.$$

Both A, B are symmetric positive definite, but for $v = (1, -0.2)^T$, $v^T BA^{-1}v = -0.14$. Thus this counterfactual composition fails operator monotonicity.

D Outcome Stability for Approximate Responses

Fix a policy π, a type ϑ, and the finite set $Z = Z(\vartheta)$ of admissible reports, and abbreviate $v(z) = v(\vartheta, z; \pi)$. At a fixed replay state u_D, let $d(z) = Q(\pi, z, \vartheta, u_D)$ be the outcome map, let z^* be a true best response, let $d^* = d(z^*)$, and define the *outcome gap*

$$\Gamma = \max_{z \in Z} v(z) - \max_{z \in Z:\, d(z) \neq d^*} v(z), \tag{31}$$

with $\Gamma = +\infty$ if no report produces another outcome.

Proposition D.1 (Strategic outcome stability). *Let $\widehat{v}$ satisfy $\sup_{z \in Z} |\widehat{v}(z) - v(z)| \leq \eta$ for some $\eta \geq 0$, and let $\widehat{z}$ be an ϵ-best response for $\widehat{v}$. Then*

$$\max_{z \in Z} v(z) - v(\widehat{z}) \leq \epsilon + 2\eta. \tag{32}$$

If moreover $\Gamma > \epsilon + 2\eta$, then $d(\widehat{z}) = d^*$.

Proof. Uniform estimation error and the approximate best-response condition give

$$\begin{aligned} v(z^*) &\leq \widehat{v}(z^*) + \eta \\ &\leq \widehat{v}(\widehat{z}) + \epsilon + \eta \\ &\leq v(\widehat{z}) + \epsilon + 2\eta. \end{aligned}$$

This proves (32). A report producing an outcome different from d^* has true regret at least Γ, so it cannot be selected under the strict gap condition. □

Multiple best responses are allowed, provided they agree on the outcome and the gap to the other outcomes is large enough. The strict threshold cannot generally be relaxed. With two reports giving different outcomes, set $v(z^*) = \epsilon + 2\eta, v(z^-) = 0, \widehat{v}(z^*) = \epsilon + \eta$, and $\widehat{v}(z^-) = \eta$. The estimate has error at most η and z^- is an ϵ-best response, yet it gives the other outcome at $\Gamma = \epsilon + 2\eta$. This can occur at arbitrarily small error and regret.

If the same error and regret bounds hold over a declared distribution of types and replay states, the pointwise result implies

$$\Pr\{d(\widehat{z}) \neq d^*\} \leq \Pr\{\Gamma \leq \epsilon + 2\eta\}.$$

A separately justified margin condition $\Pr\{\Gamma \leq t\} \leq Kt^\kappa$ yields the bound $K(\epsilon + 2\eta)^\kappa$.

Response semantics. In a policy counterfactual, the response is recomputed under the new policy with the declared tie-breaking noise. If the selection rule is unknown, the audit returns the range of outcomes over admissible responses. The proposition gives a sufficient condition for outcome stability. Whether learned negotiating agents respond approximately optimally is an empirical question it does not address.

E Experiment Protocol and Supporting Controls

E.1 Finite regimes and result semantics

The seven finite regimes are veto exactness, conjunctive mixed-layer synergy, protocol interaction with direct evidence input to the decision, a nonmonotone clause, coupled feasibility requiring a non-ancestor field, an infeasible target, and zero-cost ties. Table 2 shows the three regimes that separate the methods; the archived report contains all seven. Exhaustive enumeration supplies reference optima independently of the tested candidate ordering. Interventions retain internal assignments equal to their factual values because these can block propagation after another edit. All algorithms use the same nonnegative synthetic assignment costs and feasibility checker.

Ordered search sorts the full feasible catalogue. Single-layer search is restricted to one layer. Random search samples feasible candidates. The frozen ablation omits regeneration of intermediate state, and the ancestor-deletion ablation removes assignments that can still be needed for feasibility. Both are negative controls, evaluated against the common SCM. A null oracle margin means proven infeasibility. A null search proposal means no result at that budget, unless the certificate proves infeasibility. Invalid proposals have no defined cost gap. Relative gaps are undefined at a zero reference margin. Unconditional validity and optimal-return rates accompany conditional cost summaries in the archived report.

E.2 Affine controls

For 4,000 affine instances in dimensions 2, 5, 10, 20, we vary bounded, immutable, one-sided and blocked coordinates, including observations exactly on the threshold. Exact rational calculations find 1,145 infeasible targets. Another 1,750 cases have finite margins that are not attained. Every feasible case returns a checked successful witness within additive cost 0.01 of the infimum. Independent small-box vertex checks and the unrestricted formula test the reference implementation.

E.3 Identification and uncertainty controls

For scalar rank estimation, use $Z_0 = U$, $Z_1 = 1 + 2U$ with uniform independent U, training sample sizes 32, 128, 512, 2,048 per parent regime, and 256 held-out factual units per seed and setting. Mean counterfactual-score absolute error decreases from 0.1605 to 0.0167, with 95% seed-block interval $[0.0143, 0.0196]$ at the largest size. Three separate controls violate the assumptions: reversing the target noise rank while preserving its marginal distribution, confounding the parent with noise, and shifting the target mechanism after training. At the largest training size their errors remain approximately 0.997, 0.664 and 0.501, respectively: accurate marginal prediction leaves the cross-world coupling and its stability undetermined.

The binary response-type experiment varies compatible couplings at fixed experimental marginals. The two fair-bit models of Section 7 require the full interval $[0, 1]$. For sampled marginals we expand each empirical proportion by its simultaneous Hoeffding error radius, using failure probability 0.05 split across the two arms, then optimise the binary bound over these intervals. All 720 query/setting intervals cover the simulator truth in this run,

with mean width 0.861. Because the intervals are wide and share marginal samples, these are not 720 independent successes.

For posterior uncertainty, 2,000 held-out units follow $D = \mathbf{1}[x + U \geq 0]$ with U uniform on $[-1, 1]$. Conditioning on (x, D) gives a truncated uniform posterior. An intervention selected from this observed trace is evaluated using 256 independent posterior draws. Mean error against the analytic probability is 0.0219 and the Brier score against the held-out unit's actual counterfactual outcome is 0.1996. Reliability bins and seed-block intervals accompany the raw results.

E.4 Sequential evaluator and local checks

In the finite disclosure game of Sections 8 and 9, verification is requested only for a report meeting the announced eligibility threshold. A detected mismatch causes rejection and the penalty, and a verified matching report is approved if eligible. Expected utility is approval benefit 5 minus effort $k|z - \vartheta|$ and expected mismatch penalty. All three reports are evaluated, so best responses have zero regret within this finite set. The complete best-response set is retained, and a separate indexed draw selects among ties.

An early report intervention regenerates the request, reply and decision. Noise keys include seed, episode, mechanism and time, so an unrelated call cannot consume another mechanism's draw. This coupling is part of the simulator, not a property assumed of external APIs. The policy comparison changes the eligibility threshold from 1 to 2, separating fixed disclosure from a regenerated best response. The archive retains complete example traces and the finite factorial design.

Complete local equation checks reject 12,000/12,000 single-node corrupted records in the finite suite, including terminal and skip-input corruptions. The check takes the mechanisms, boundary inputs, latent state and intervention as trusted. It verifies the evaluation, not the authenticity of the record or the fidelity of the causal model.

E.5 Finite checks of the posterior results

The check script uses seed 20260915, rational probabilities and floating-point logarithms. It checks 40 conditioned veto SCMs with seven eligible actions, including 45,312 intervention/state replays, 360 greedy comparisons with exhaustive optima, 35 Set Cover encodings across 4,480 subsets, and counterexamples to reverse-direction and robust-envelope submodularity. It also checks the sampled-corollary implication on the uniform-error event and finite strategic-stability instances. These are consistency checks of the implementation against the proofs.

E.6 Posterior search in random veto models

The experiment draws 20 seed blocks of 50 instances, with block seeds $20261009 + b$, using the script `posterior_search_experiment.py`. Each instance is a veto SCM with seven intervenable nodes and a terminal decision, and 24 latent states. In each state, every node draws its active parents among earlier nodes independently with probability 0.3 and has base value one with probability 0.82. The first state has all base values one, so an approval exists. Prior weights are integers from 1 to 9. A factual state is drawn uniformly among approving states, and the trace records nodes 0, 2 and the decision. The posterior is the prior restricted to compatible states. The layers are nodes $\{0, 1, 2\}$, $\{3, 4\}$ and $\{5, 6\}$, consistent with the topological order. Action costs are integers from 1 to 9. Exact joint and single-layer optima enumerate all 128 action sets. Greedy search is run twice: stopped exactly at Q, which is reported in the paper and lies outside the guarantee, and with slack $1/20$, for which the bound of the greedy theorem is asserted in every feasible case. The script also asserts that single-layer search is exact at every known state. Seed-block percentile intervals use 2,000 bootstrap resamples with seed 20261010. Results, configuration and source hashes are written to `review/posterior-search.json`. The generator settings are inherited from the finite theorem checks and are not calibrated to a lending population.

E.7 Excluded evidence and next evaluations

Earlier exploratory lending sweeps used policies and intervention protocols that do not match the present definitions, so they are not reported. A study with hosted language-model agents needs fixed mechanisms and prompts, a declared estimand, matched financial rules, retained failures and independent validation after selection. Usefulness to human principals and empirical strategic behaviour need separate evaluation.

Archived lending illustrations. Earlier single-borrower lending grids are retained as historical artifacts, but excluded from these comparisons: the rule and LLM scripts use different margin thresholds (5% and 10%), and the saved artifacts do not establish a matched fixed-expense intervention protocol. A loan/revenue threshold is affine in the plotted coordinates; with fixed expenses, principal-only 24-month amortisation and margin floor q, the reference approval frontier would instead be

$$R \geq \max\{E_0/(1-q),\ E_0 + 0.625L,\ L/0.55\}.$$

This is piecewise affine, not a ratio-induced hyperbola. A nearest rejected grid point gives only a grid/slice upper bound under a declared cost. A single hosted response per input at temperature zero establishes neither repeatability nor a zero margin. Matched-policy LLM evaluation requires new, separately logged runs and independent validation.

F Satisfiability and Threshold Phenomena

This outlook develops the satisfiability connection in the conference supplement. The posterior-search theorems do not depend on a random-model threshold claim.

Satisfiability formulations. At a known latent state with Boolean mechanisms, the decision under an intervention is a Boolean circuit in the admissible assignments. Minimum cost satisfiability provides a constraint-solver formulation, as in counterfactual and abductive explanations of fixed classifiers [21, 22]. In the veto class, the cheapest known-state flip is simply the cheapest active ancestor. For an explicit posterior on states $\mathbf{u}_1, \ldots, \mathbf{u}_k$ with positive weights w_j, write $R_j = R(\mathbf{u}_j)$. Each state contributes the positive clause

$\bigvee_{v \in R_j} x_v$. The posterior margin at confidence Q is the optimum of

$$\begin{aligned} &\min_{x,y} \sum_{v \in I} c_v x_v, \\ &y_j \le \sum_{v \in R_j} x_v \quad (j = 1, \dots, k), \\ &\sum_j w_j y_j \ge Q \sum_j w_j, \qquad x, y \in \{0,1\}. \end{aligned} \tag{33}$$

The auxiliary variable y_j marks a covered state. The inequalities suffice because one may set $y_j = 1$ for every covered state. This is an integer-programming formulation and can also be encoded using weighted partial MaxSAT with cost and coverage constraints. It avoids explicitly listing the $2^{|I|}$ interventions, although it does not remove worst-case hardness. Our experiment in Section 9.2 enumerates 2^7 sets and does not benchmark large constraint solvers.

At $Q = 1$, every positive-weight state must be covered. With no action budget, choosing every action satisfies every nonempty positive clause. An empty R_j makes confidence one impossible. Thus a nontrivial feasibility threshold for this class needs a restricted catalogue, such as $\sum_v c_v x_v \le B$, or another stated constraint. It is not the satisfiability problem for random clauses with mixed signs.

With succinct Boolean mechanisms and independent fair latent bits, choosing an intervention before the random state gives an existential-random problem. This includes E-MAJSAT, an $\mathrm{NP}^{\mathrm{PP}}$-complete problem [27]. The auditor's order is existential then random. A state-aware oracle chooses after observing the state, which explains its lower cost in Proposition 6.1.

Many latent states at a fixed action dimension. Suppose the posterior is uniform on k independently sampled states, whose ancestor sets are independent copies of $R \sim \mu$. Fix a finite whitelist I and L layers, allow all its subsets, and impose no action budget in this paragraph. Put

$$p_i = \mu(R \cap \mathbf{V}_i \neq \varnothing), \qquad p = \mu(R \neq \varnothing).$$

The largest pure-layer success is attained by selecting a whole layer, and the largest joint success by selecting I. The law of large numbers therefore gives pure feasibility with probability tending to one for $Q < \max_i p_i$ and to zero for $Q > \max_i p_i$. Joint feasibility has the corresponding boundary at p. A window in which only mixed audits succeed exists when $p > \max_i p_i$. Equivalently, for every layer there must be positive probability that another layer hits R while that layer does not. Non-nesting of just two hitting events is insufficient when a third layer can cover their union.

For fixed L, Hoeffding's inequality bounds the probability that any of these $L + 1$ empirical success rates differs from its mean by more than t by $2(L + 1)e^{-2kt^2}$. Thus the uncertainty band in the *confidence level* is of order $k^{-1/2}$ at fixed error probability. This is not a scaling-window theorem in clause density with growing action dimension. The experiment in Section 9.2 uses conditioned, nonuniform posterior weights and a fixed number of prior states, so it illustrates pure-versus-mixed feasibility without testing this asymptotic statement.

Confidence-one audits as covering problems. Let $\mathcal{X}_B$ be a nonempty finite catalogue of interventions within a declared budget. For each state $\mathbf{u}$, let $Y(\mathbf{u})$ be the members of $\mathcal{X}_B$ that fail at that state. A common intervention succeeding at all k states exists exactly when

$$\bigcup_{j=1}^{k} Y(\mathbf{u}_j) \neq \mathcal{X}_B.$$

If the states are independent and identically distributed and each candidate has positive failure probability, the first k for which the union covers $\mathcal{X}_B$ is a finite cover time T. Let $\mathcal{L}$ be the last uncovered subset and $h(A)$ the expected time for a fresh independent sequence to cover A. Aldous's theorem gives

$$\mathrm{Var}(T) \le K\, \mathbb{E}[T]\, \mathbb{E}[h(\mathcal{L})]$$

for a universal constant K [3]. Along a family of such problems, $\mathbb{E}[T] \to \infty$ and $\mathbb{E}[h(\mathcal{L})]/\mathbb{E}[T] \to 0$ imply $T/\mathbb{E}[T] \to 1$ in probability. Establishing this condition for a specified random audit model is a separate task. If one candidate never fails, the failure sets cannot cover the catalogue and this finite-cover argument does not apply.

The same covering perspective appears in random satisfiability. Recent preprints by Carenini give scaling-window bounds for sparse monotone covering problems [7] and claim, using a criterion of Abbe and Montanari [1], a limiting satisfiability threshold for every fixed clause width at least three [6]. Earlier results established sharpness [14] and the threshold for sufficiently large clause width [11]. These results concern their specified random ensembles. Whether budgeted random audit models satisfy the required structural hypotheses remains open. For $Q < 1$, a candidate may fail on some posterior mass, giving a partial-satisfaction problem rather than ordinary covering of the candidate catalogue.

Continuous interventions. Consider the homogeneous special case in which state j succeeds exactly when $a_j^T \delta > 0$, with unrestricted $\delta \in \mathbb{R}^d$. A common intervention exists if and only if the a_j lie in an open half-space through the origin. For independent, centrally symmetric random vectors in general position, Wendel's theorem gives probability

$$2^{-(k-1)} \sum_{i=0}^{d-1} \binom{k-1}{i},$$

with a transition near $k = 2d$ as the dimension grows [49]. Nonzero score margins, one-sided changes and cost budgets lead to a different feasibility problem. Statistical dimension supplies a useful approach when the audit can be cast in a random conic model meeting the geometric assumptions of Amelunxen et al. [4]. It does not by itself determine a threshold for arbitrary constrained negotiations. Locating these transitions and comparing pure with mixed search are questions for future work.